Supporting Guidance on the Compliance Algorithm for the Urban Waste Water Treatment Directive (91/271/EEC)

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Introduction

This document introduces a compliance algorithm designed to systematically evaluate wastewater data against regulatory requirements, with reference to the Urban Waste Water Treatment Directive 91/271/EEC (UWWTD). The purpose of this algorithm is to make the compliance process transparent, structured, and accessible. It is intended for a broad professional audience — including scientists, legal experts, and project managers — regardless of their specific technical expertise in wastewater engineering.

Notation

Category	Acronym	Meaning / Explanation
Area Types	SA	Sensitive Area
	CSA	Catchment of Sensitive Area
	A54	Sensitive Area applying Article 5(4)
	A58	Sensitive Area applying Article 5(8)
	NA	Normal Area
	LSA	Less Sensitive Area
Water Types	ES	Estuary
	FW	Freshwater
	CW	Coastal Water
	LC	On land (catchment of coastal water)
	LF	On land (catchment of freshwater and/or estuary)
Compliance Status	C	Compliant / Calculated (context-dependent)
	NC	Not Compliant
	NI	No Information
	PD	Pending Deadline[1]
	NR	Not Relevant
	QC	Questionable Compliance [2]
	AddQC	Additional Questionable Compliance [3]
Measurements / Calculations	p.e.	Population equivalent
	BOD5	Biological Oxygen Demand over 5 days
	COD	Chemical Oxygen Demand
	E	Estimated
	M	Measured

[1] If a country has been granted a deadline extension for implementing the Directive (see pages 6, 8 and 22 for details)
[2] See page 24 for details
[3] See page 24 for details

Overview of the Compliance Algorithm for the Urban Wastewater Treatment Directive (UWWTD) (91/271/EEC)

The level of treatment required depends on the type of receiving area — freshwater, estuarine, or coastal — and on whether it is classified as a sensitive area (due to susceptibility to eutrophication, its use for drinking water production, or designation under other Directives) or as a less sensitive area (open marine waters).

The algorithm is organised into a sequence of decision trees, applied step by step:

1. **Determine required treatment**
   The first step is to calculate the required level of treatment for each urban wastewater treatment plant (UWWTP). This calculation is based on two factors:
   - the size of the agglomeration, and
   - the type of receiving area.

   See Algorithms:
   - Algorithm 1a. UWWTP: Treatment required
   - Algorithm 1b. UWWTP: Treatment required (Sensitive Areas)
   - Algorithm 1c. UWWTP: Treatment Required (Catchment of Sensitive Areas)

2. **Assess treatment in place**
   Next, the treatment in place at each Urban Wastewater Treatment Plant (UWWTP) must be assessed:
   - First, verify that the treatment installed matches the required treatment calculated in Step 1.
   - Then, check whether the performance results for all parameters required are reported as passing.

   Only if both conditions are met — the correct treatment in place *and* satisfactory performance — the treatment plant is considered compliant.

   See Algorithms:
   - Algorithm 2a. UWWTD: Treatment and Performance Compliance (except more stringent treatment)
   - Algorithm 2b. UWWTD: Treatment and Performance Compliance (more stringent treatment)

3. **Country awarded a delay**
   If a country has been granted a deadline extension for implementing the Directive, an exception applies:
   - Treatment plants that are not yet compliant are excluded from further compliance calculations.
   - These plants are classified as “Pending Deadline (PD)”.

   See Algorithm 3. UWWTD: Treatment and Performance Compliance (Correction for Transitional Period)

4. **Agglomeration compliance (Article 3)**
   The next step is to calculate the compliance of the agglomeration with Article 3, which concerns wastewater collection.

   See Algorithm 4. Agglomeration: Article 3 Compliance

5. **Agglomeration Excluded**
   Because of limitations in the quality of reported data or other special cases, some agglomerations must be excluded from the subsequent compliance calculations.

   These special cases may be one of the following:
   - Not enough information: all end points with at least one *"?"*
   - 100% treated in Individual and Appropriate Systems (IAS)
   - Agglomeration <10000pe which discharges in waterbody of type LC
   - Agglomeration <10000pe which discharges in waterbody of type CW

   See Algorithm 5. Agglomeration: Article 4, 5, 6 Compliance Exclusion

6. **Agglomeration compliance (Article 4, 5, and 6)**
   To determine the compliance of an agglomeration with the three relevant articles (4, 5, and 6). Since many agglomerations have **multiple treatment plants**, the first step is to calculate the compliance of each individual treatment plant with the requirements of these three articles.

   See Algorithm 6. Agglomeration: Plant Compliance with Article 4, 5, and 6

7. **Aggregate the compliance of all treatment plant for an Agglomeration**
   Once the compliance of each treatment plant has been determined, the compliance of the agglomeration can be calculated by aggregating the compliance of all treatment plants connected to that agglomeration. This aggregation is done separately for each article.

   See Algorithms:
   - Algorithm 7. Agglomeration: Article 4 only
   - Algorithm 8. Agglomeration: Article 6 only
   - Algorithm 9. Agglomeration: Article 5 only

8. **Overall compliance**
   To determine the overall compliance of an agglomeration:
   - Use the compliance results for each individual article calculated in previous steps.
   - Apply the general rule: if any single article is non-compliant, the agglomeration as a whole is considered non-compliant.

   See Algorithm 10. Agglomeration: Overall Compliance

9. **Legal compliance**
   To determine the legal compliance for each article (4, 5, and 6), apply the hierarchical compliance rule, which defines how the compliance of individual treatment plants and agglomerations influences the legal status of each article.

See Algorithm 11. Agglomeration: Legal Compliance of Articles 4, 5, and 6
If a situation is non-compliant but the deadline has not yet passed, it is shown as "Not Relevant", indicating that it is non-compliant but still within the allowed timeframe. Meanwhile, the platform internally records the calculation as "Pending Deadline" to enable administrators to easily track and verify these cases.

Definition of Legal Compliance

Compliance with the main requirements of the Directive is assessed per agglomeration. Full compliance requires meeting three conditions:

1. **Collection of wastewater** via sewers or Individual and Appropriate Systems (IAS) – necessary for compliance with Article 3, and also a prerequisite for compliance with Articles 4 and 5 (hierarchical approach).
2. **Secondary treatment** of collected wastewater, i.e., adequate treatment level and performance, is required for compliance with Article 4.
3. **More stringent treatment than secondary treatment**, i.e., higher treatment level and performance – generally required for compliance with Article 5, i.e. required for agglomerations >10,000 p.e. discharging into sensitive areas or their catchment, or collectively for the group of all agglomerations >2,000 p.e. discharging into a sensitive area of type art5(4) where nutrient removal performance (N and P) must reach at least 75% of the total load of all treatment plants.

Hierarchical Approach to Compliance

The legal compliance assessment follows a hierarchical approach:

- Non-compliance with collection obligations (Article 3) automatically implies non-compliance with treatment obligations (Articles 4 and, when applicable, 5).
- An agglomeration that fails to meet secondary treatment standards cannot be considered compliant with Article 5.

Agglomerations may be partially compliant:

- For example, an agglomeration may adequately collect wastewater (compliant with Article 3) but treat it inadequately (non-compliant with Article 4 or Articles 4 and 5).
- Agglomerations failing to collect wastewater properly (non-compliant with Article 3) cannot be considered compliant with any other Directive requirements.

For non-compliant agglomerations, the portion of wastewater that meets compliance is not included in overall compliance rate calculations (regional, national, or EU levels).

Tolerances

The total wastewater load of an agglomeration is considered compliant even if[4]:

- ≤2% (and ≤2,000 p.e.) of the wastewater load does not comply with Article 3, and

[4] For more information regarding the tolerance, please check: https://circabc.europa.eu/ui/group/65764c73-4a57-45dc-8199-473014cf65bf/library/ea692049-6cc8-464d-a0a5-663bc11b98d6/details

- ≤1% (and ≤2,000 p.e.) of the collected wastewater load does not comply with Article 4 and/or Article 5.

Calculation of Compliance Rates

- **Full compliance rate** at the Member State level is based on the wastewater load of all agglomerations in full compliance.
- **Article 3 compliance rate** (collection) is calculated from the load of agglomerations fully compliant with collection obligations, relative to the total load of reported agglomerations (≥2,000 p.e.).
- **Article 4 compliance rate** (secondary treatment) is calculated from the load of agglomerations fully compliant with secondary treatment obligations, relative to the total load of reported agglomerations with obligations under Article 4. This includes agglomerations discharging into sensitive or non-sensitive areas:
    - ≥2,000 p.e. discharging into freshwater or estuaries, or
    - ≥10,000 p.e. discharging into coastal waters.
- **Article 5 compliance rate** (more stringent treatment) is calculated from the load of agglomerations fully compliant with nitrogen and/or phosphorus removal obligations, relative to the total load of agglomerations subject to Article 5 (≥10,000 p.e. discharging into sensitive areas).

Specific case of Article 5(4): In principle an area subject to Article 5(4) should only be considered compliant if all collection and secondary treatment situated in that area is compliant with the relevant requirements and if the reductions of N and P are at least 75%. In this case, the compliance should be assessed at the level of agglomerations for Article 3 and Article 4, and at the level of areas subject to Article 5(4) for Article 5.

Full compliance with the Directive is calculated on the basis of the wastewater load of the agglomerations in full compliance (Article 3 and, where applicable, Articles 4 and 5) in reference to the total load of all the reported agglomerations.

Definition of Distance to Target (DTT)

The term distance to target represents the remaining effort required for a Member State to achieve compliance with the Directive for each requirement (per Article) at the reporting level. This is a conceptual tool not included in the Directive itself.

It indicates how far a Member State is from meeting the goals for collecting and treating wastewater. The rules used to define this concept are as follows:

- **Distance to target – collection**: the total load of wastewater that should be collected according to the Directive but is not yet collected.
- **Distance to target – treatment:** the total load of collected wastewater that is not treated at all or not treated adequately. Additionally, if the load of wastewater that is not collected or addressed through individual systems (IAS) exceeds 2%, this load is also considered to

require treatment[5]. Inadequately treated wastewater refers to discharged water that does not meet the Directive's quality requirements, e.g., BOD_5, COD, and, where applicable, total phosphorus and/or total nitrogen concentrations.

Collection target: All wastewater covered by the Directive (Article 3) is collected. The distance to target for collection shows how much wastewater is still uncollected or lost during transport.

Treatment target: All wastewater covered by the Directive (Articles 4 and/or 5) is adequately treated (meeting required treatment level and performance). The distance to target for treatment shows how much of the collected and uncollected wastewater is still not adequately treated.

Generally, the overall treatment target is that all wastewater from agglomerations >10,000 p.e. discharging into sensitive areas is adequately treated with more stringent treatment than secondary treatment. The distance to target figure for more stringent treatment indicates the amount of collected and uncollected wastewater not yet treated to this level.

Key Considerations

- Distance to target values cannot be compared directly to legal compliance values.
- They provide a more nuanced view of Directive implementation because they take into account the compliant fraction of treated wastewater in agglomerations that are not fully legally compliant.

[5] Please refer to https://circabc.europa.eu/ui/group/65764c73-4a57-45dc-8199-473014cf65bf/library/ea692049-6cc8-464d-a0a5-663bc11b98d6/details for more information

Compliance Algorithm

Each of the decision trees are explained in more detail, providing an overview, pseudocode, and the decision tree. For internal record keeping, the "Exit Leafs" of the algorithm are logged (the blue boxes in the decision tree and pseudocode).

Algorithm 1a. UWWTP: Treatment required

Overview

First decision step in determining what wastewater treatment level is legally "required" for a given plant, based on:

- How big the plant's connected agglomerations are (population equivalent load)
- What type of water body the discharge goes into (e.g., freshwater, coastal, sensitive area)
- Special deadlines for compliance to one or more articles of the 1991 UWWTD (Articles 3, 4, 5)
- Whether the plant is active and connected to an agglomeration

Simplified Logic

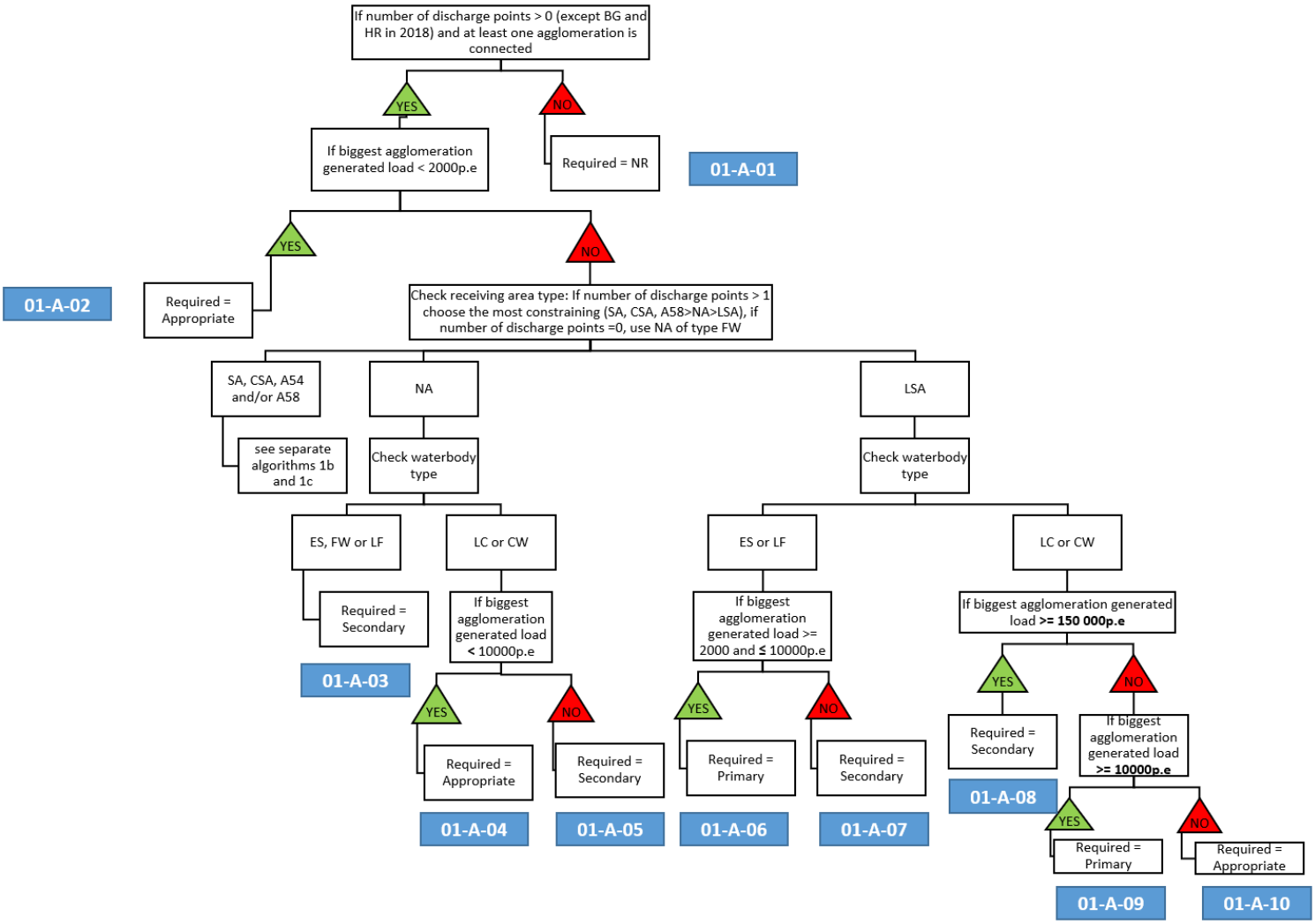
1. Low load case
    - If load < 2000 →
        - result_required = Appropriate
        - exit_leaf = 01-A-02
2. Receiving area dependent cases

    a. Receiving area in {SA, CSA, A54, A58, A58523, A523, A5854}
    - Call Algorithm 1b

    b. Receiving area = NA
    - If water in {ES, FW, LF} →
        - result_required = Secondary
        - exit_leaf = 01-A-03
    - Else if water in {LC, CW} →
        - If load < 10000 → result_required = Appropriate, exit_leaf = 01-A-04
        - Else → result_required = Secondary, exit_leaf = 01-A-05

    c. Receiving area = LSA
    - If water in {ES, LF} →
        - If 2000 ≤ load ≤ 10000 → result_required = Primary, exit_leaf = 01-A-06
        - Else → result_required = Secondary, exit_leaf = 01-A-07
    - Else if water in {LC, CW} →
        - If load ≥ 150000 → result_required = Secondary, exit_leaf = 01-A-08
        - Else if load ≥ 10000 → result_required = Primary, exit_leaf = 01-A-09
        - Else → result_required = Appropriate, exit_leaf = 01-A-10

Decision Tree



Algorithm 1b. UWWTP: Treatment required (Sensitive Areas)

Overview

Algorithm 1b determines the required wastewater treatment level for a treatment plant when the most restrictive type of receiving area (TypeOfReceivingArea) is one of SA, CSA, A54, A58, or related codes.

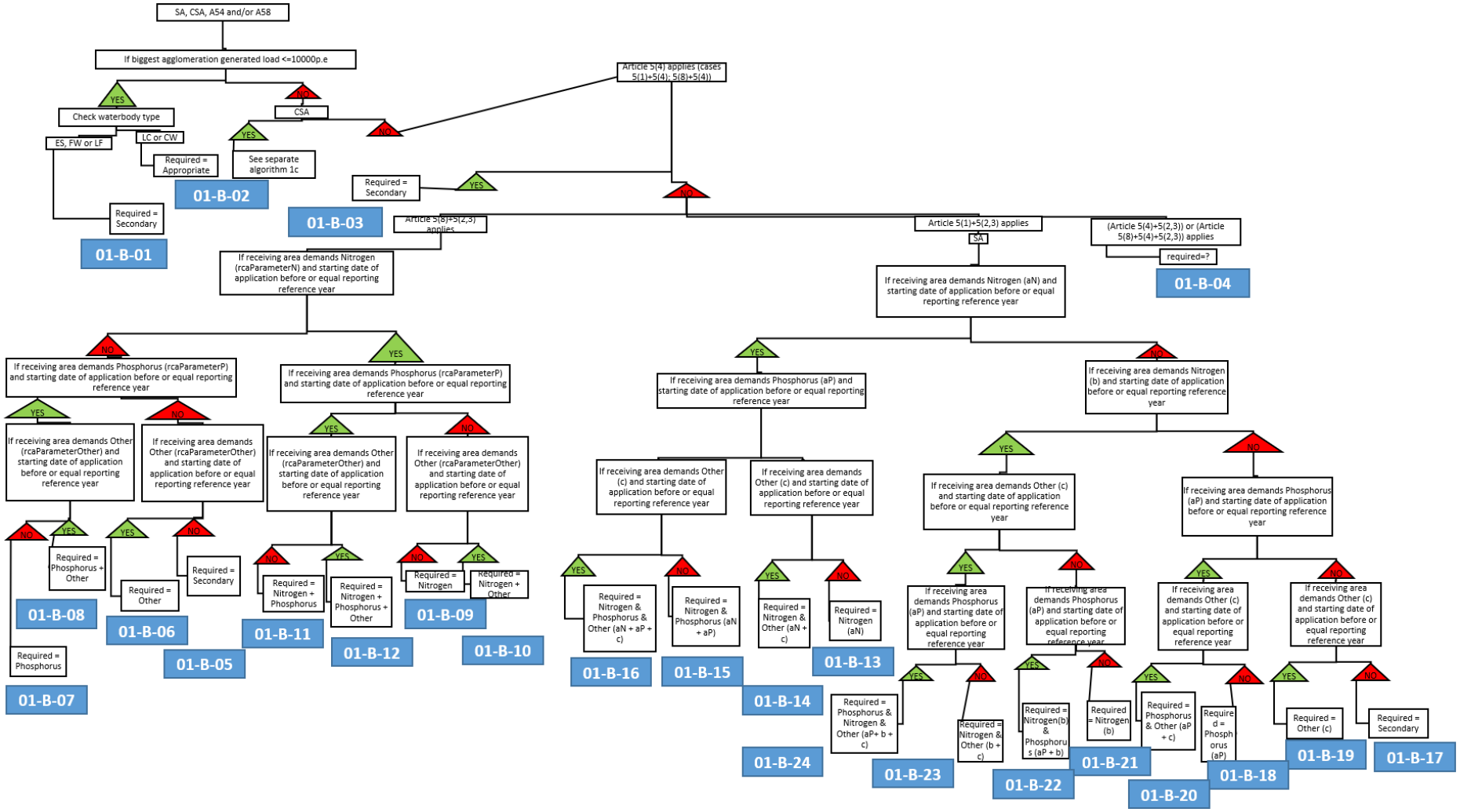
It considers:

- The size of the agglomeration (biggestLoad).
- The type of water body at the discharge point.
- Which articles apply (articleApplies).
- Receiving area parameters for Nitrogen, Phosphorus, and other treatments.
- Dates related to reporting and start of application.

Simplified Logic:

1. Check article_applies:
    - Only continues if /A523/ appears in the string. Otherwise, nothing happens.
2. Check dcp_type:
    - If CSA, the algorithm delegates to Algorithm 1c.
    - If SA, A523, or A58523, it processes Algorithm 1b rules.
3. Check if rca_a_nitro is 1 and start date <= reporting date:
    - Determines which combination of nutrients (aN, aP, c) is required.
    - Sets a result string and exit leaf code based on these combinations.
4. If the Nitrogen condition is not met:
    - Checks combinations of rca_b, rca_a_phos, and rca_c for “secondary” cases.
    - Assigns the corresponding result string and exit leaf code.
5. Update plant record:
    - Stores the calculated result and exit_leaf using a placeholder function update_algorithm_plant.

Decision Tree



Algorithm 1c. UWWTP: Treatment Required (Catchment of Sensitive Areas)

Overview

Determine the nutrient requirements for a plant based on the nitrogen and phosphorus flags for the receiving area (Eutrophication: rcaANitro, rcaAPhos; Quality of water for drinking: rcaB)

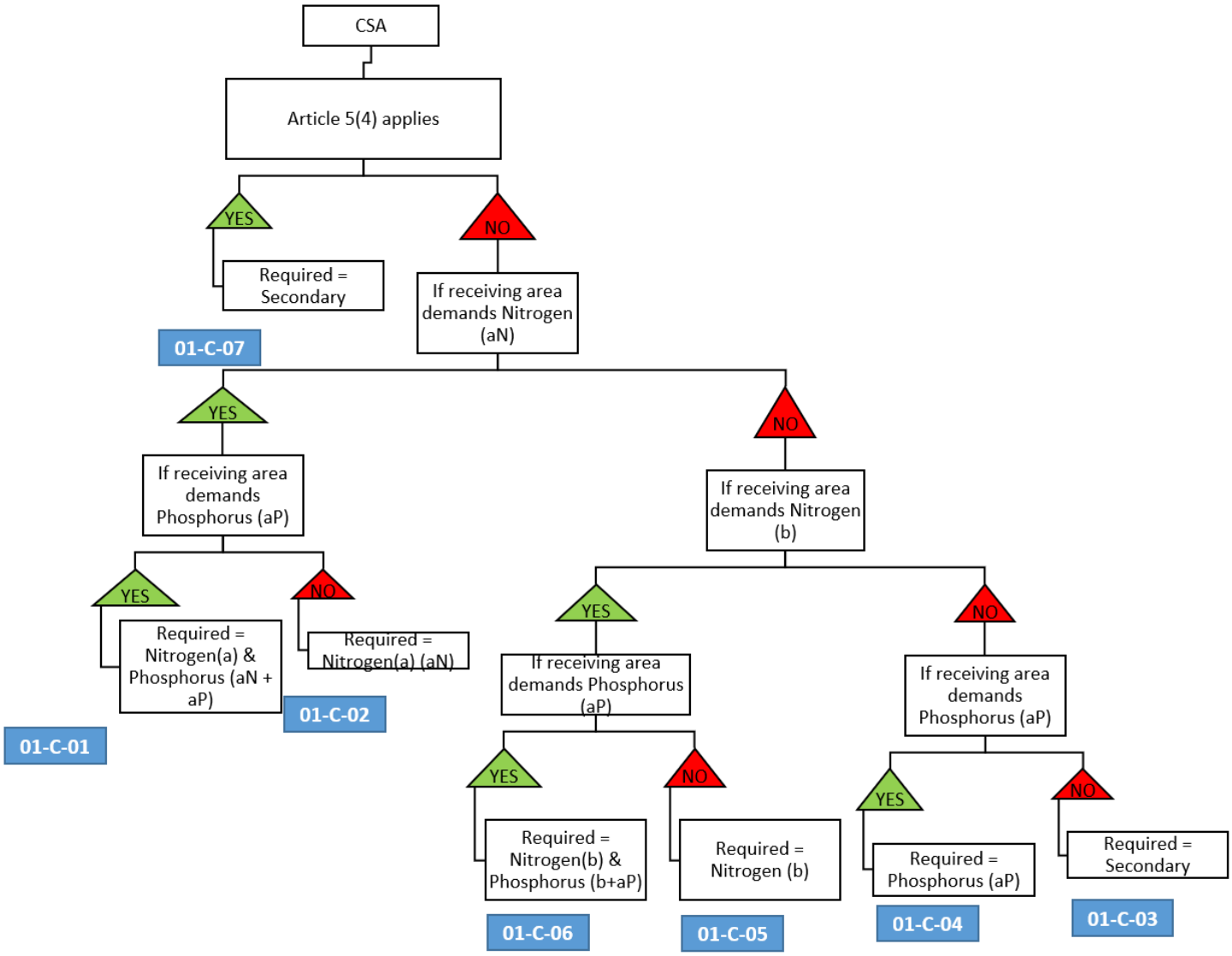
Inputs

- rcaANitro → Does the primary receiving area demand Nitrogen?
- rcaAPhos → Does it demand Phosphorus?
- rcaB → Secondary Nitrogen demand.

Simplified Logic:

1. Check Article 5(4):
    - If it applies, the result is automatically "Secondary" and exit leaf "01-C-07".
    - If it does not apply, we proceed to check nutrient demands.
2. Check primary Nitrogen (rcaANitro):
    - If rcaANitro = 1:
        - If rcaAPhos = 1: Required = (aN+aP), Exit = 01-C-01
        - Else: Required = (aN), Exit = 01-C-02
    - If rcaANitro = 0:
        - Use combinations of rcaB and rcaAPhos to determine:
            - rcaB=0 & rcaAPhos=0 → Secondary, 01-C-03
            - rcaB=0 & rcaAPhos=1 → (aP), 01-C-04
            - rcaB=1 & rcaAPhos=0 → (b), 01-C-05
            - rcaB=1 & rcaAPhos=1 → (aP+b), 01-C-06

Decision Tree



Algorithm 2a. UWWTD: Treatment and Performance Compliance (except more stringent treatment)

Overview

Determines whether a treatment plant is compliant, non-compliant, or indeterminate based on:

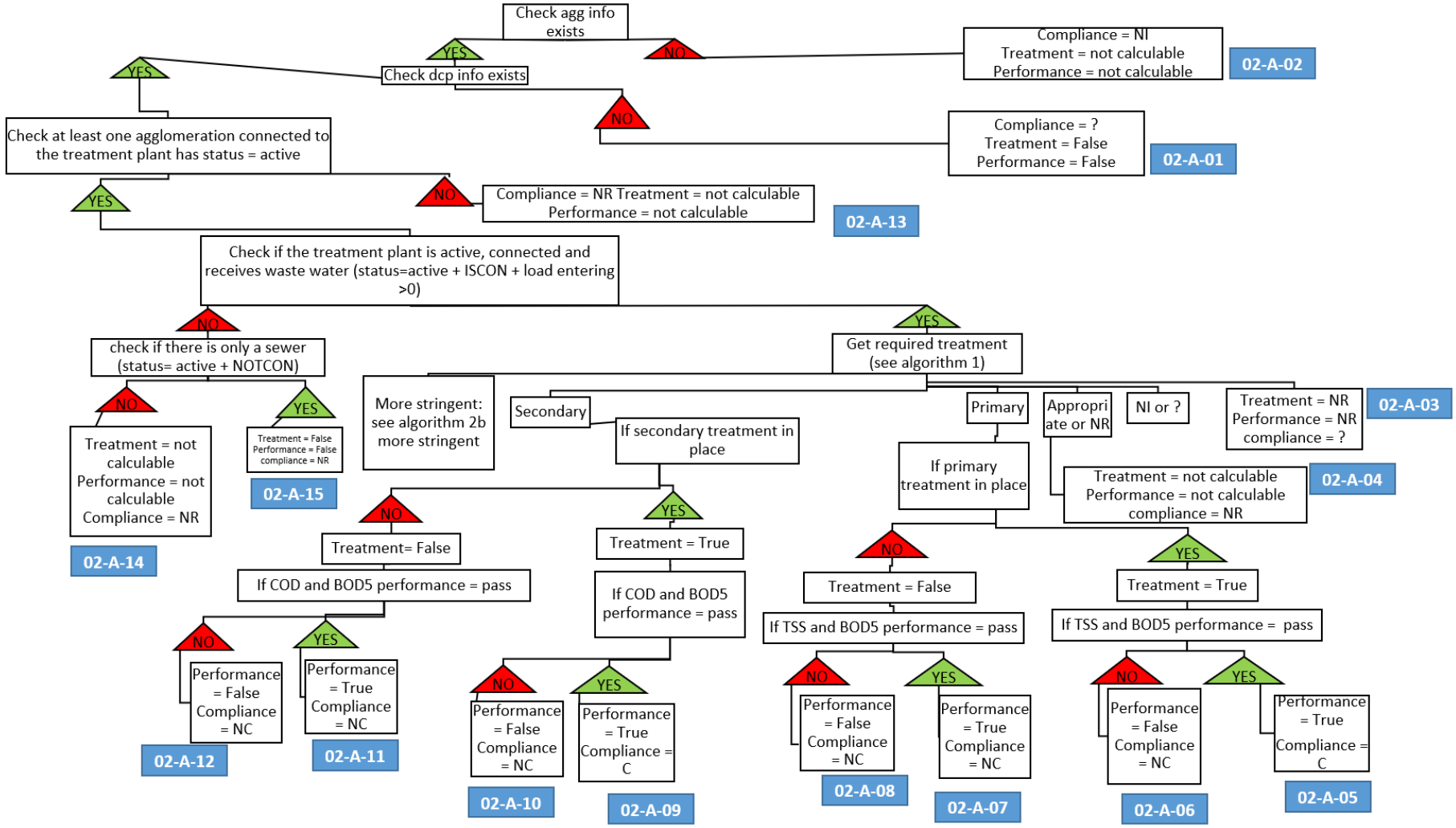
- the existence of discharge points,
- whether connected agglomerations are active,
- whether the plant itself is active and connected,
- whether the plan is active but not connected (sewer with no treatment plant),
- and the required treatment level (NR, Appropriate, Primary, Secondary, or More Stringent).

It then checks whether the actual treatment in place and performance (BOD5, TSS, COD) meet the required standard.

Simplified Logic:

1. Check discharge points (dcp_exists)
    - If no discharge points exist → compliance is unknown (?).
    - If info missing → compliance is *not information* (NI).
2. Check if there is an active connected agglomeration (agg_exists_active)
    - If none → result is "NR / not calculable".
3. Check plant status (active_and_connected, active_and_not_connected)
    - If active & connected → evaluate treatment requirements.
    - If active but not connected → "NR / False / False".
    - If not active or not connected → "NR / not calculable".
4. Check required treatment (result_required)
    - "?" or "NI" → Compliance unknown (?), NR required.
    - "Appropriate" or "NR" → Compliance NR, not calculable.
    - "Primary" →
        - If primary treatment installed:
            - Performance good (TSS & BOD5 pass) → Compliant (C).
            - Otherwise → Non-compliant (NC).
        - If not installed:
            - Performance good → NC.
            - Otherwise → NC.
    - "Secondary" →
        - If secondary treatment installed:
            - Performance good (COD & BOD5 pass) → Compliant (C).
            - Otherwise → NC.
        - If not installed:
            - Performance good → NC.
            - Otherwise → NC.
    - "More stringent" → Passes control **to Algorithm 2b.**

Decision Tree



Algorithm 2b. UWWTD: Treatment and Performance Compliance (more stringent treatment)

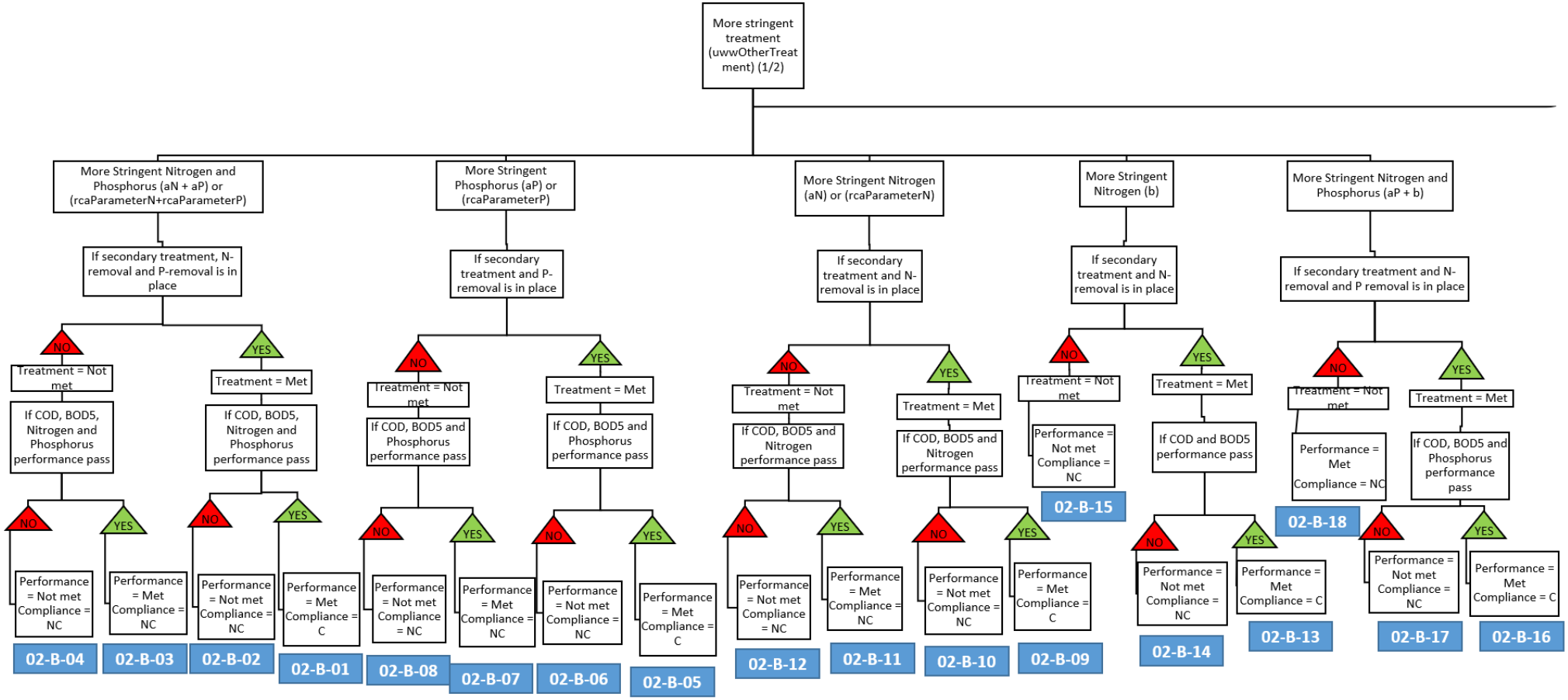
Overview

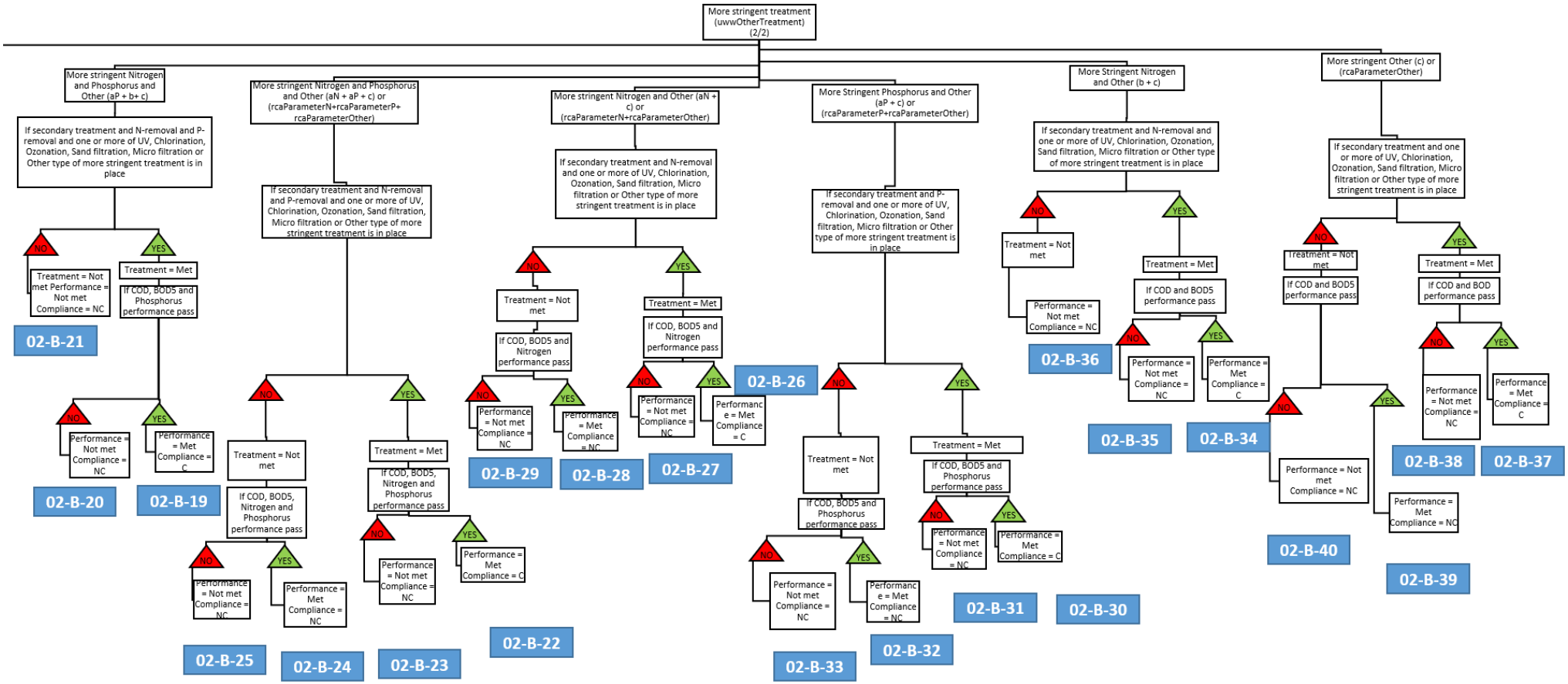
Wastewater treatment compliance based on the type of required treatment, presence of actual treatment, and performance metrics. It assigns compliance results (C = Compliant, NC = Non-Compliant), treatment and performance flags.

1. Look up the "Required" treatment type from Algorithm 1 (result_required).
    - Could be things like "phosphorus", "nitrogen", "other", "(aN+aP)", "(aP+b)", "(aP+b+c)", etc.
2. For each required case, it runs a branch of the decision tree:
    - Does the plant have the right infrastructure? (e.g., nitrogen removal installed?)
    - Did the plant's performance tests pass for the relevant pollutants?
3. Set three outputs:
    - result_treatment → whether the plant has the right equipment.
    - result_performance → whether the plant's monitoring shows compliant performance.
    - result_compliance → "C" (compliant) or "NC" (non-compliant).

Decision Trees

The decision tree is split into 2 for legibility, but can be thought of as one tree with many branches.




More stringent treatment (uwwOtherTreatment) (2/2)
More stringent Nitrogen and Phosphorus and Other (aP + b+ c)
If secondary treatment and N-removal and P-removal and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met Performance = Not met Compliance = NC
Treatment = Met
If COD, BOD5 and Phosphorus performance pass
02-B-21
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-20
02-B-19
More stringent Nitrogen and Phosphorus and Other (aN + aP + c) or (rcaParameterN+rcaParameterP+ rcaParameterOther)
If secondary treatment and N-removal and P-removal and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met
Treatment = Met
If COD, BOD5, Nitrogen and Phosphorus performance pass
If COD, BOD5, Nitrogen and Phosphorus performance pass
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = NC
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-25
02-B-24
02-B-23
02-B-22
More stringent Nitrogen and Other (aN + c) or (rcaParameterN+rcaParameterOther)
If secondary treatment and N-removal and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met
Treatment = Met
If COD, BOD5 and Nitrogen performance pass
If COD, BOD5 and Nitrogen performance pass
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = NC
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-29
02-B-28
02-B-27
02-B-26
More Stringent Phosphorus and Other (aP + c) or (rcaParameterP+rcaParameterOther)
If secondary treatment and P-removal and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met
Treatment = Met
If COD, BOD5 and Phosphorus performance pass
If COD, BOD5 and Phosphorus performance pass
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = NC
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-33
02-B-32
02-B-31
02-B-30
More Stringent Nitrogen and Other (b + c)
If secondary treatment and N-removal and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met
Treatment = Met
If COD and BOD5 performance pass
Performance = Not met Compliance = NC
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-36
02-B-35
02-B-34
More stringent Other (c) or (rcaParameterOther)
If secondary treatment and one or more of UV, Chlorination, Ozonation, Sand filtration, Micro filtration or Other type of more stringent treatment is in place
NO
YES
Treatment = Not met
Treatment = Met
If COD and BOD5 performance pass
If COD and BOD performance pass
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = NC
NO
YES
Performance = Not met Compliance = NC
Performance = Met Compliance = C
02-B-40
02-B-39
02-B-38
02-B-37

Algorithm 3. UWWTD: Treatment and Performance Compliance (Correction for Transitional Period)

Overview

This is the transitional period correction step. It takes the output of Algorithm 2 (treatment/performance compliance) and adjusts it depending on deadlines in the 1991 UWWTD:

- Article 4 → requires secondary treatment.
- Article 5 → requires more stringent treatment (nutrients or advanced).

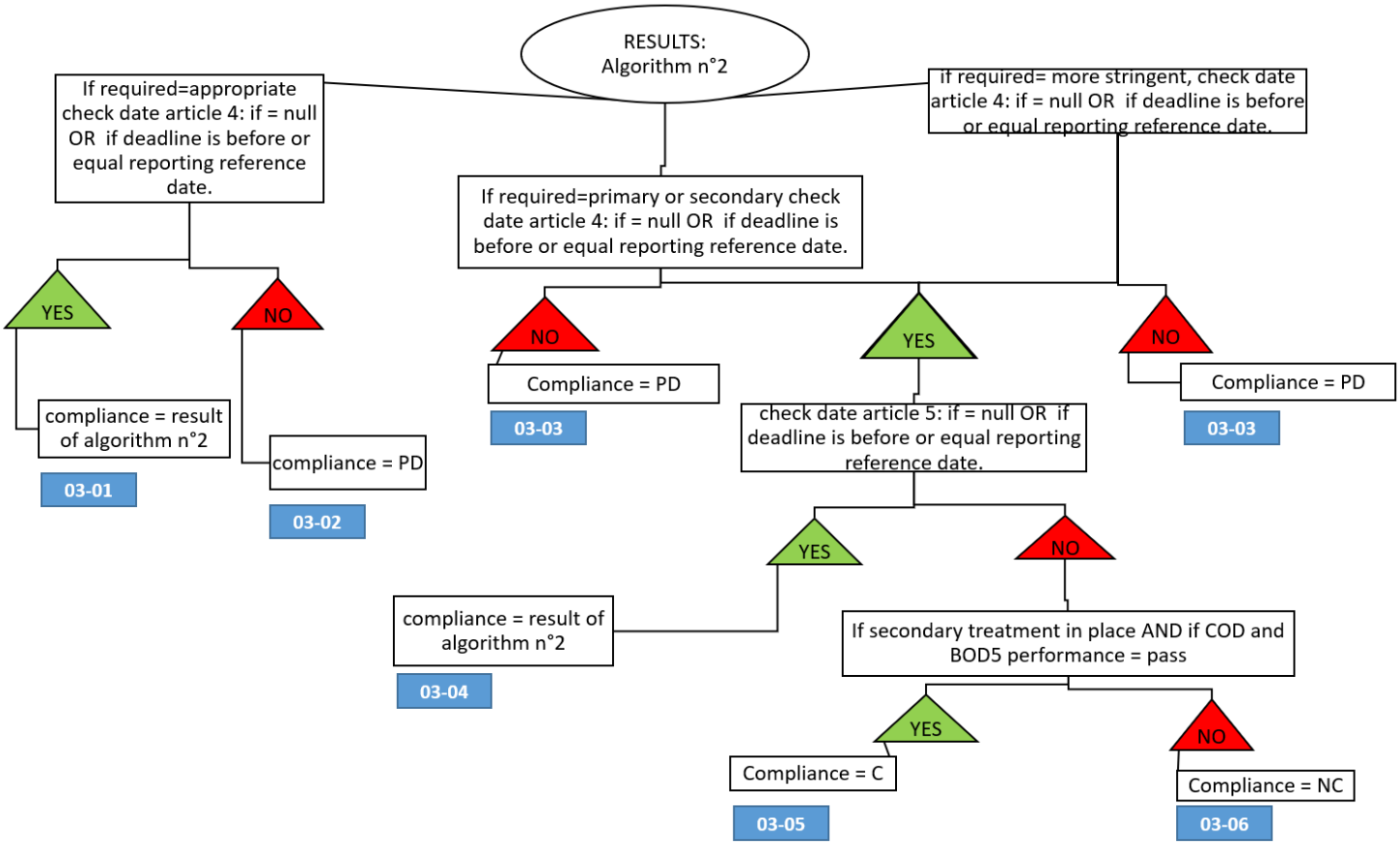
Some countries had derogations/delays. If the reporting date (repSituationAt) is before the legal deadline, then a non-compliant plant may temporarily be classified as "Pending Deadline (PD)" instead of "Non-Compliant".

Algorithm 3's role is to take compliance from Algorithm 2 and adjust it to C / NC / PD, depending on deadlines and minimal checks.

Simplified Logic:

1. If required = "Appropriate"
    - Deadline of Article 4 passed or unknown → compliance = result from Algorithm 2 (alg3_exit_leaf = 03-01).
    - Deadline of Article 4 not yet passed → compliance = PD (alg3_exit_leaf = 03-02).
2. If required = primary, secondary, or more stringent
    - Deadline of Article 4 passed or unknown → check Article 5 deadline:
        - Deadline of Article 5 passed or unknown → compliance = result from Algorithm 2 (alg3_exit_leaf = 03-04).
        - Deadline of Article 5 not yet passed → check plant treatment:
            - Secondary treatment present and COD & BOD5 pass → compliance = C (alg3_exit_leaf = 03-05).
            - Otherwise → compliance = NC (alg3_exit_leaf = 03-06).
    - Deadline of Article 4 not yet passed → compliance = PD (alg3_exit_leaf = 03-03).

Decision Tree



Algorithm 4. Agglomeration: Article 3 Compliance

Overview

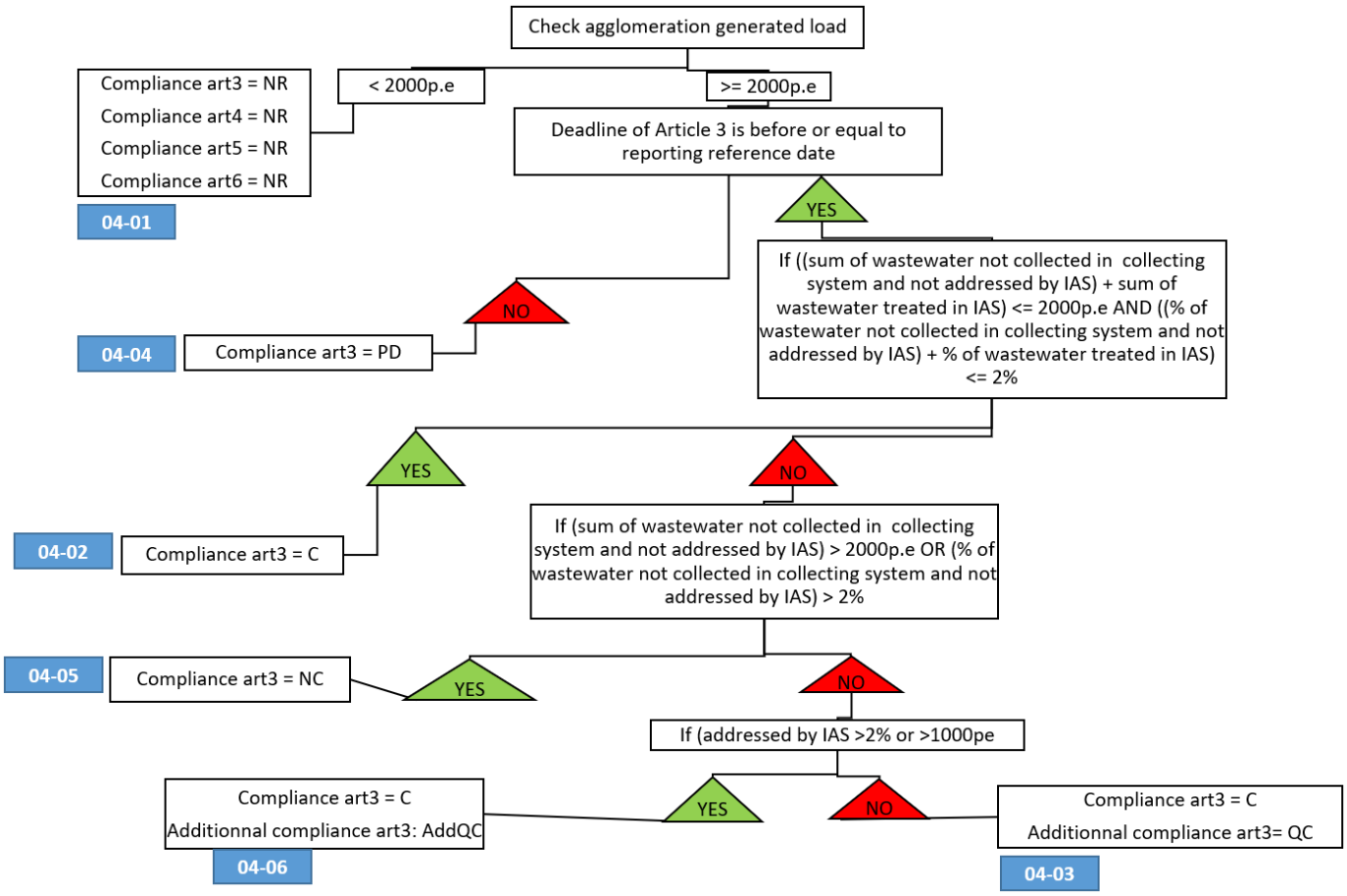
This procedure checks whether an agglomeration ≥ 2000 p.e. has an adequate wastewater collection system by deadline date.

- If <2000 p.e. → not relevant (NR).
- If deadline not reached → pending (PD).
- If deadline reached → several branches:
  - Very small untreated load → compliant (C).
  - Too much untreated load → non-compliant (NC).
  - In-between → compliant but flagged with extra QC (when a load is addressed by IAS (<=2% and <=1000p.e.)) or AddQC (significant load is addressed by IAS (>2% or >1000p.e. of the generated load of the agglomeration) depending on IAS.

Simplified Logic:

1. Low load case
   - If aggGenerated < 2000 → compliance = NR for Art 3, 4, 5, 6 (alg4_exit_leaf = 04-01).
2. Agglomeration ≥ 2000
   - Deadline of Article 3 passed or unknown (aggDateArt3 <= repSituationAt):
     1. If ((wastewater not collected + wastewater treated in IAS) ≤ 2000 p.e. AND (% not collected + % treated in IAS) ≤ 2%) → compliance = C (alg4_exit_leaf = 04-02).
     2. Else:
        - If (% not collected > 2 OR sum not collected > 2000 p.e.) → compliance = NC (alg4_exit_leaf = 04-05).
        - Else:
          - If (wastewater treated in IAS > 2% OR sum treated in IAS > 1000 p.e.) → compliance = C + AddQC (alg4_exit_leaf = 04-06).
          - Else → compliance = C + QC (alg4_exit_leaf = 04-03).
   - Deadline of Article 3 not yet passed → compliance = PD (alg4_exit_leaf = 04-04).

Decision Tree



Algorithm 5. Agglomeration: Article 4, 5, 6 Compliance Exclusion

Overview

Filters agglomerations based on connected plants and their treatment compliance, setting Art. 4–6 compliance accordingly, and delegates further processing to Algorithm 6 if conditions are not conclusive. It also calls Algorithm 10 and 11 when all plants are NR/NI with required "Appropriate treatment".

Simplified Logic:

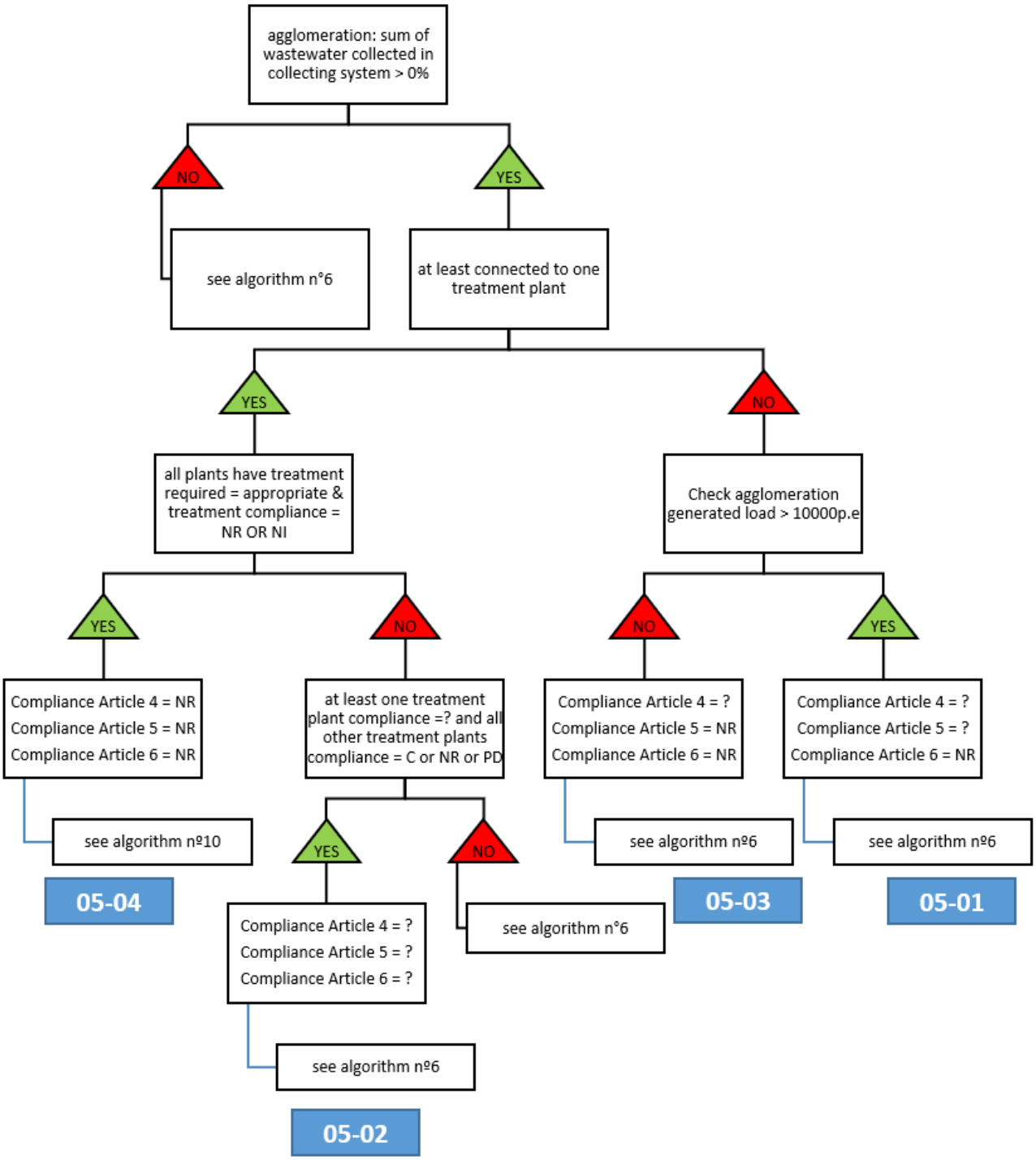
Some wastewater collected (sum_percent_wastewater_collected > 0)

1. Checks if plants are connected:

   2a. At least one plant connected
   - Checks if all connected plants require Appropriate treatment and have compliance = NR or NI:
     - If yes:
       - Art. 4, 5, 6 compliance = NR.
       - Calls Algorithm 10 and Algorithm 11 for additional processing.
     - If not:
       - Checks if at least one plant has compliance = "?" and the rest are C/NR/PD:
         - If yes → Art. 4, 5, 6 compliance = "?" (unknown), exit leaf 05-02.
         - Otherwise → calls Algorithm 6 to continue processing.

   2b. No plants connected
   - Two sub-cases based on aggGenerated:
     - aggGenerated > 10000 → Art. 4 & 5 = "?", Art. 6 = NR, exit 05-01.
     - aggGenerated < 10000 → Art. 4 = "?", Art. 5 & 6 = NR, exit 05-03.

Decision Tree



Algorithm 6. Agglomeration: UWWTP Compliance for Article 4, 5, and 6

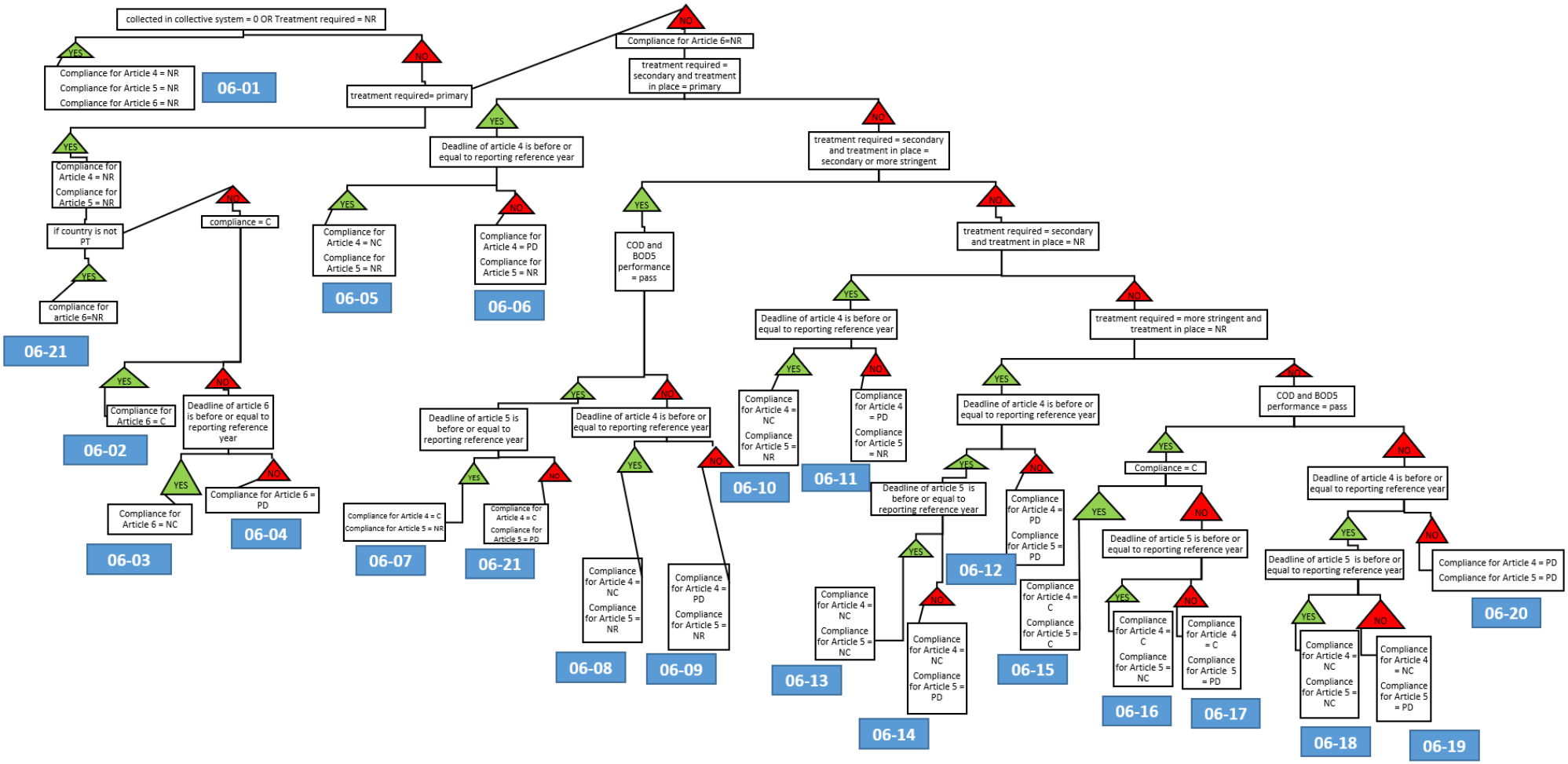
Overview

Determines the compliance status (NR, NC, PD, C) based on treatment requirements, treatment in place, COD/BOD performance, deadlines, and country-specific rules.

Simplified Logic

1. No wastewater collected or treatment not required (NR)
    - Directly mark compliance NR for all articles (exit leaf 06-01).
2. Treatment required is Primary
    - For all countries except PT all compliance = NR (06-21)
    - If compliance = C → Art6 = C, Art4/5 = NR (06-02)
    - Else → check if Portugal then Art.6 deadline vs reporting year (C or NC or PD) (06-02 / 06-04)
3. Treatment required is Secondary
    - Primary treatment in place → check Art.4 deadline (NC / PD) (06-05 / 06-06)
    - Secondary or stringent treatment in place → check COD/BOD performance → check Art.4 deadline (C, NC, PD) (06-07 → 06-09)
    - Treatment not in place → fallback to deadline checks and more stringent rules (06-10 → 06-11)
4. Treatment required is More stringent treatment for N and/or P and/or other parameters
    - More stringent treatment in place → check COD/BOD performance → if pass check Art.5 deadline (C, NC, PD) (06-15 →06-17)
    - if fail: check Art.4 and Art.5 deadlines (C, NC, PD) (06-18→06-20)
    - Treatment not in place → fallback to deadline checks for Art.4 and Art.5 and rules (06-12 → 06-14)

Decision Tree



Algorithm 7. Agglomeration: Article 4 only

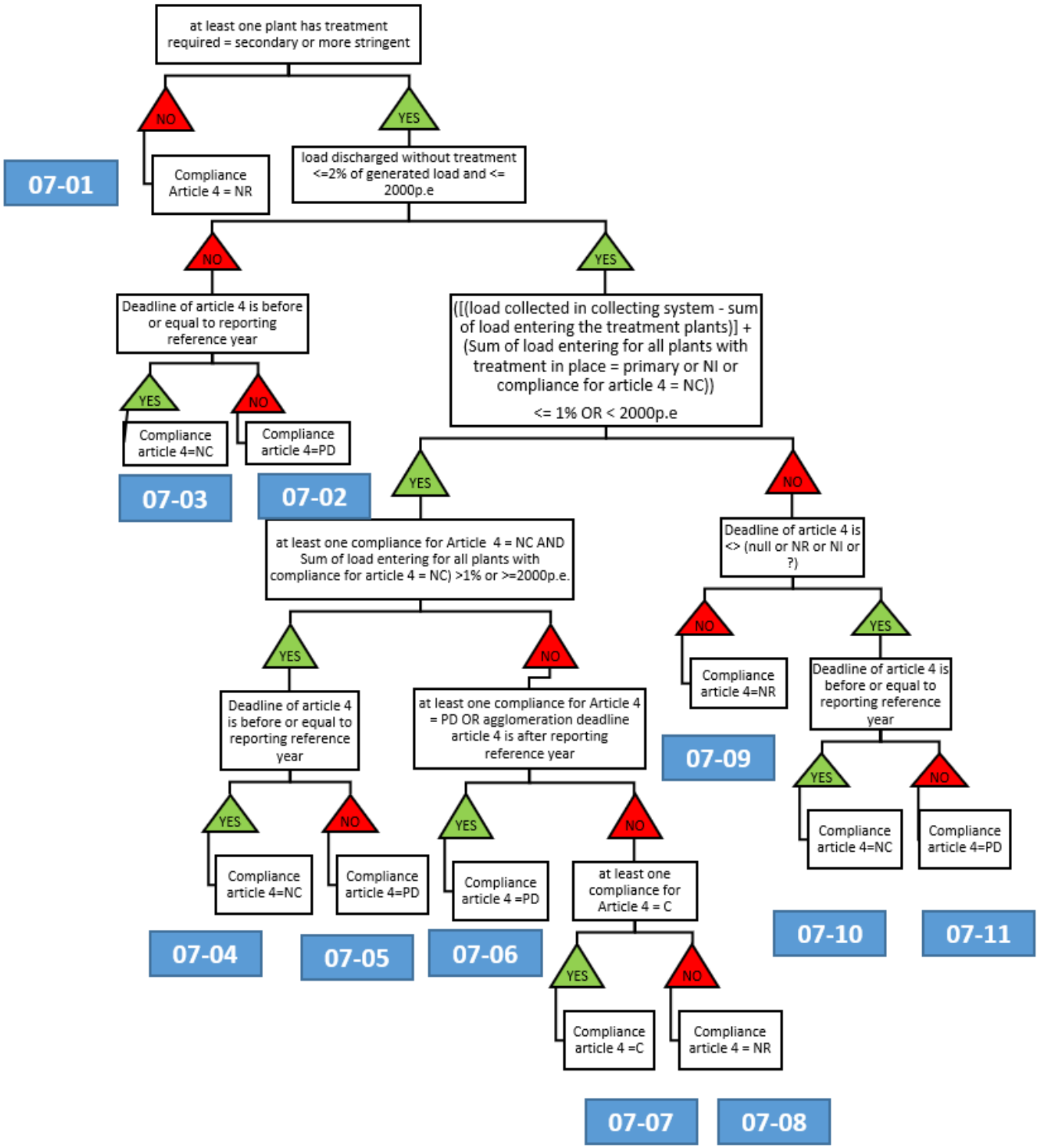
Overview

Compliance of an agglomeration with Article 4 of the 1991 urban wastewater treatment directive. The compliance depends on the aggregation of compliance statuses of individual treatment plants and on the load discharged without treatment.

Simplified Logic

1. No secondary/more stringent treatment required → NR (exit 07-01).
2. Load discharged without treatment ≤ 2% and ≤ 2000 p.e. → check load entering stations and individual plant compliance.
    - Depending on deadlines and number of plants with NC, PD, C → set agglomeration compliance (NC, PD, C).
3. Load discharged without treatment > 2% or > 2000 p.e. → check Article 4 deadline vs reporting year → NC or PD.
4. No plant satisfies conditions → fallback to NR.

Decision Tree



Algorithm 8. Agglomeration: Article 6 only

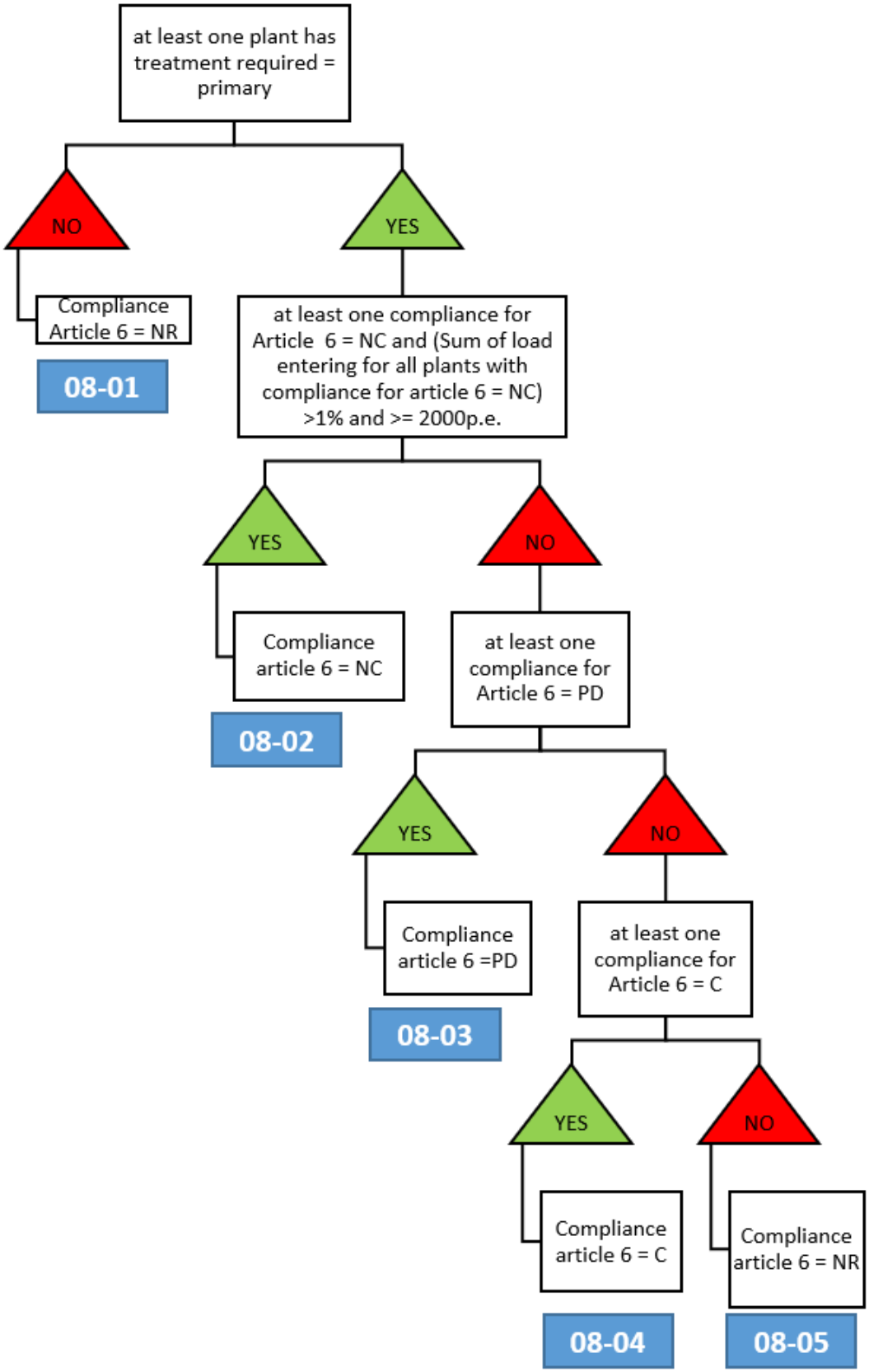
Overview

Calculate compliance of an agglomeration with Article 6 of the 1991 urban wastewater treatment directive. Compliance depends on treatment type at connected plants and load entering plants.

Simplified Logic:

1. No primary treatment required → NR (exit 08-01).
2. At least one primary treatment plant exists:
    - If any plant compliance = NC and load entering NC plants > 1% of collected load and ≥ 2000 p.e. → NC (exit 08-02).
    - Else, if any plant compliance = PD → PD (exit 08-03).
    - Else, if any plant compliance = C → C (exit 08-04).
    - Else → NR (exit 08-05).

Decision Tree



Algorithm 9. Agglomeration: Article 5 only

Overview

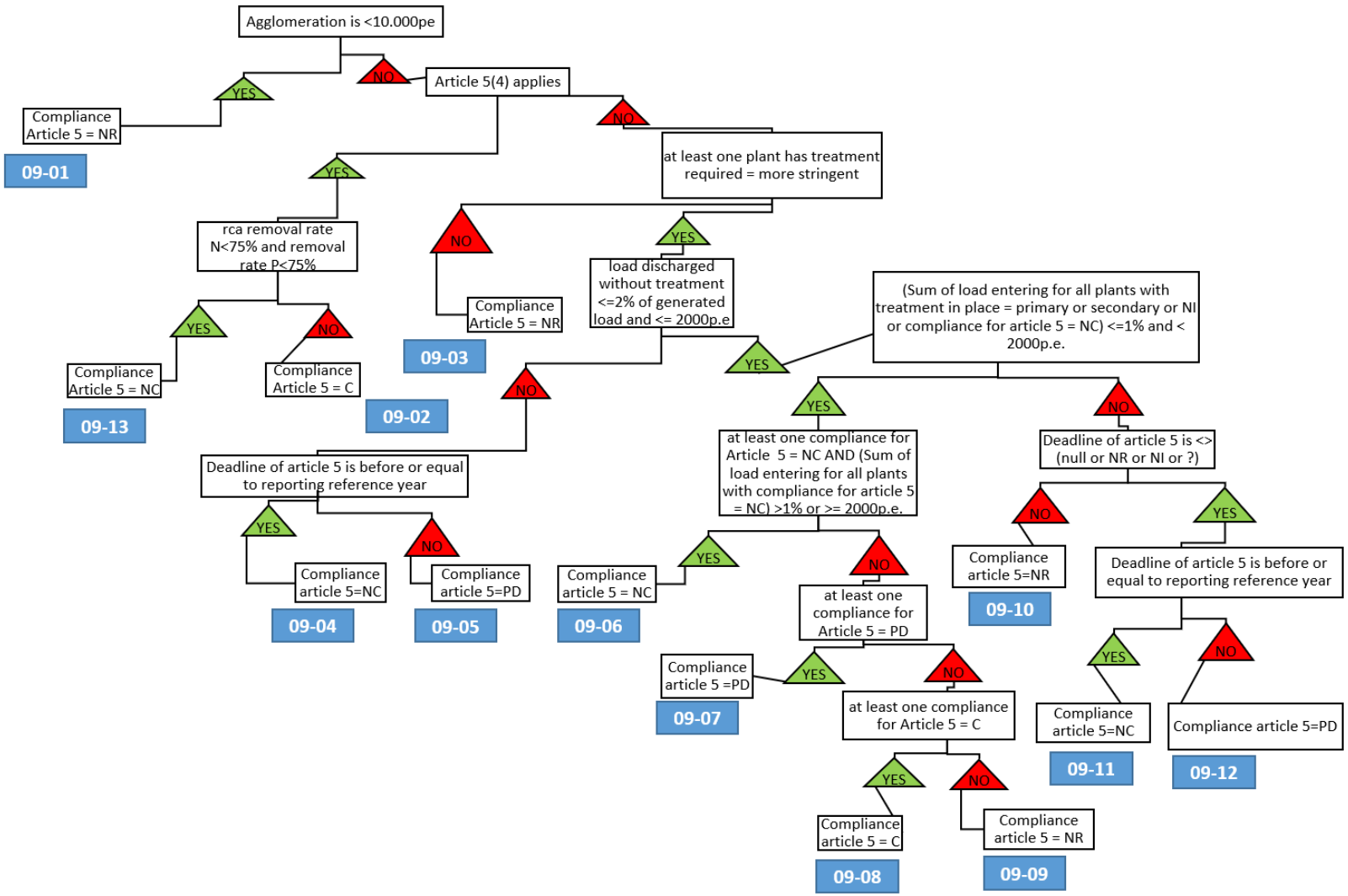
Determine compliance of an agglomeration with Article 5 of the 1991 urban wastewater treatment directive, which focuses on nitrogen (N) and phosphorus (P) removal but also other more stringent parameters such as disinfection or specific treatments required for the purpose of the implementation of other EU Directives, and treatment stringency. Compliance depends on:

- Agglomeration size.
- Applicability of Article 5(4).
- RCA removal rates.
- Load discharged without treatment.
- Load entering treatment plants.
- Treatment level of connected plants.
- Deadline for Article 5 relative to reporting year

Simplified Logic:

1. Agglomeration < 10,000 pe → NR (exit 09-01).
2. Agglomeration ≥ 10,000 pe:
    - If Art.5(4) applies and N & P removal ≥ 75% → C (exit 09-02).
    - Else, if Art.5(4) applies and N or P removal < 75% → NC (exit 09-13).
    - Else, if no more stringent treatment required → NR (exit 09-03).
    - Else, check load discharged without treatment:
        - If ≤ 2% and ≤ 2000 p.e.:
            - If any plant NC and load entering NC > 1% & ≥ 2000 → NC (exit 09-06).
            - Else, if any plant PD → PD (exit 09-07).
            - Else, if any plant C → C (exit 09-08).
            - Else → NR (exit 09-09).
        - Else, check Article 5 deadline vs reporting date:
            - aggDateArt5 ≤ repSituationAt → NC (exits 09-04 or 09-11).
            - aggDateArt5 > repSituationAt → PD (exits 09-05 or 09-12).

Decision Tree



Algorithm 10. Agglomeration: Overall Compliance

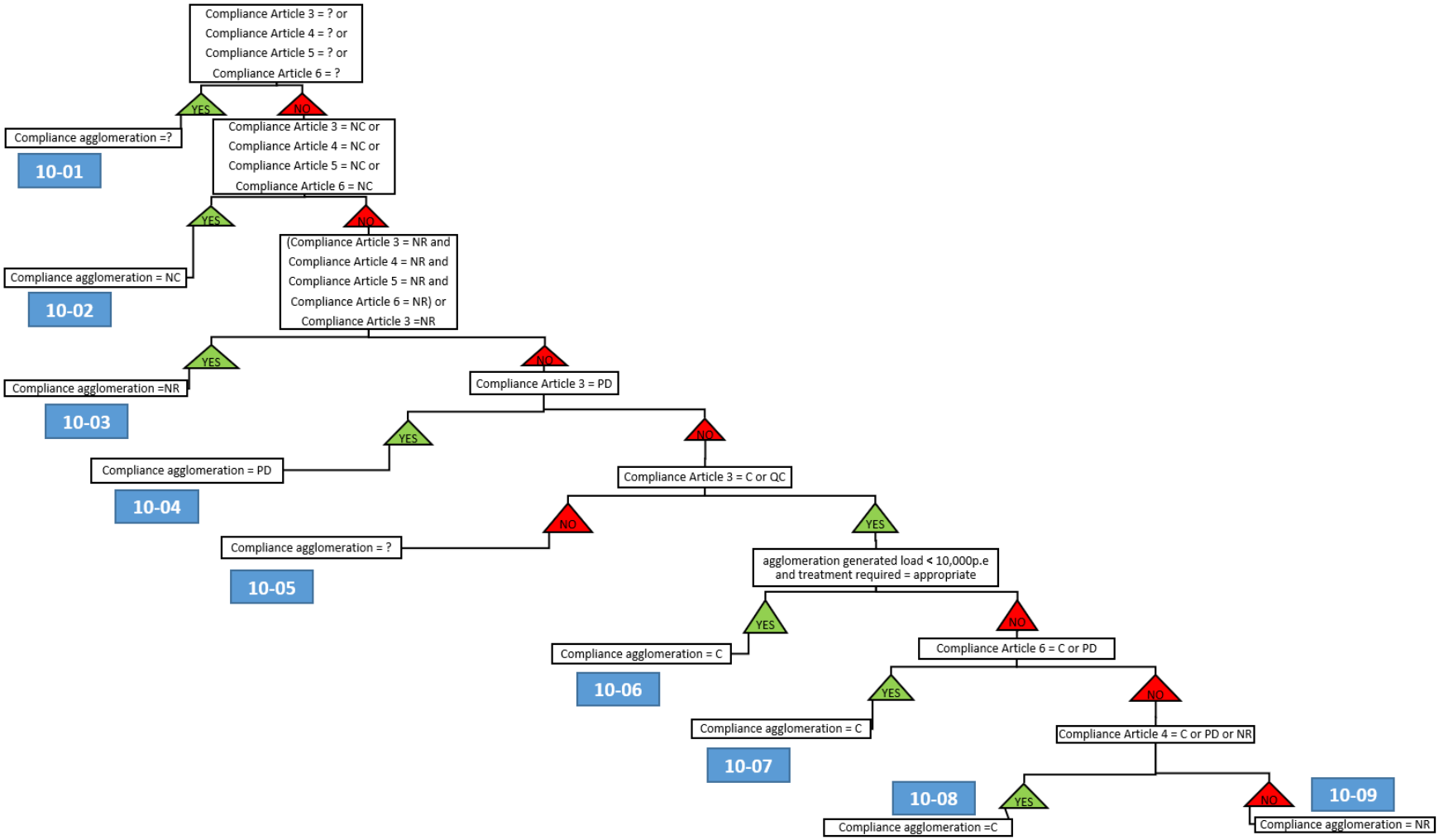
Overview

Overall compliance of an agglomeration based on the compliance results of Articles 3, 4, 5, and 6. This is a hierarchical logic, where certain conditions override others.

Simplified Logic:

1. Check for NULLs → Raises error if any article compliance is NULL.
2. Check for unknown (?) → Overall compliance = ? (exit 10-01).
3. If any Article = NC → Overall compliance = NC (exit 10-02).
4. If Article 3 = NR → Overall compliance = NR (exit 10-03).
5. If Article 3 = PD → Overall compliance = PD (exit 10-04).
6. If Article 3 not in (C, QC) → Overall compliance = ? (exit 10-05).
7. If aggGenerated < 10,000 & at least one plant with appropriate treatment → C (exit 10-06).
8. If Article 6 = C or PD → Overall compliance = C (exit 10-07).
9. If Article 4 = C, PD, or NR → Overall compliance = C (exit 10-08).
10. Otherwise → Overall compliance = NR (exit 10-09).

Decision Tree



Algorithm 11. Agglomeration: Legal Compliance of Articles 4, 5, and 6

Overview

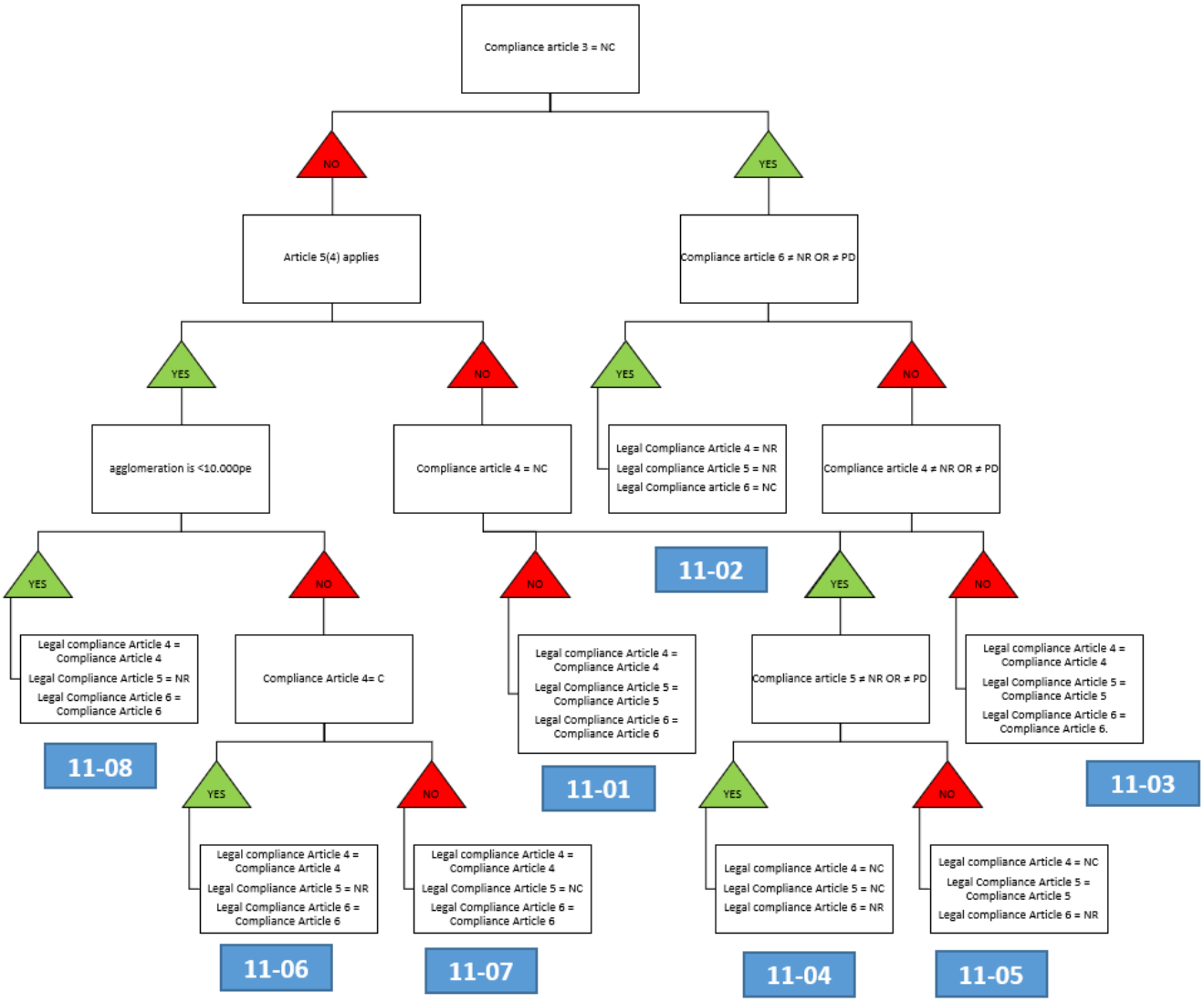
Determine the legal compliance for Articles 4, 5, and 6 at the agglomeration level, based on:

- Compliance of Article 3 (result_compliance_art3)
- Compliance of Articles 4, 5, and 6 (result_compliance_art4/5/6)
- Agglomeration characteristics (aggGenerated, np_54)

Simplified Logic:

1. Check NULLs → Raises error if any input compliance is NULL.
2. If Article 3 = NC
    - Check Article 6:
        - If Article 6 ≠ NR and ≠ PD → Art4 = NR, Art5 = NR, Art6 = NC (exit 11-02)
        - Else check Article 4:
            - If Article 4 ≠ NR and ≠ PD → check Article 5:
                - If Article 5 ≠ NR and ≠ PD → Art4 = NC, Art5 = NC, Art6 = NR (exit 11-04)
                - Else → Art4 = NC, Art5 = Article5, Art6 = NR (exit 11-05)
            - Else → No change (Art4/5/6 stay same) (exit 11-03)
3. If Article 3 ≠ NC
    - Check np_54 (Article 5(4) applies):
        - If yes → check aggGenerated < 10000:
            - If yes → Art4 = Compliance art4, Art5 = NR, Art6 = Compliance art6 (exit 11-08)
            - Else → check Article 4:
                - If Article 4 = C → Art4 = Compliance art4, Art5 = NR, Art6 = Compliance art6 (exit 11-06)
                - Else → Art4 = Compliance art4, Art5 = NC, Art6 = Compliance art6 (exit 11-07)
        - If no → check Article 4:
            - If Article 4 = NC → check Article 5:
                - If Article 5 ≠ NR and ≠ PD → Art4 = NC, Art5 = NC, Art6 = NR (exit 11-04)
                - Else → Art4 = NC, Art5 = Article5, Art6 = NR (exit 11-05)
            - Else → No change (exit 11-01)

Decision Tree



Annexe I: PseudoCode of the Compliance Algorithms

Algorithm 1a. UWWTP: Treatment required

```
# Low load case
elif load < 2000:
    result_required = "Appropriate"
    exit_leaf = "01-A-02"

# Receiving area dependent cases
else:
    if receiving_area in {"SA", "CSA", "A54", "A58", "A58523", "A523", "A5854"}:
        algorithm_1b()  # Call Algorithm 1b

    elif receiving_area == "NA":
        if water in {"ES", "FW", "LF"}:
            result_required = "Secondary"
            exit_leaf = "01-A-03"
        elif water in {"LC", "CW"}:
            if load < 10000:
                result_required = "Appropriate"
                exit_leaf = "01-A-04"
            else:
                result_required = "Secondary"
                exit_leaf = "01-A-05"

    elif receiving_area == "LSA":
        if water in {"ES", "LF"}:
            if 2000 <= load <= 10000:
                result_required = "Primary"
                exit_leaf = "01-A-06"
            else:
                result_required = "Secondary"
                exit_leaf = "01-A-07"
        elif water in {"LC", "CW"}:
            if load >= 150000:
                result_required = "Secondary"
                exit_leaf = "01-A-08"
            elif load >= 10000:
                result_required = "Primary"
                exit_leaf = "01-A-09"
            else:
                result_required = "Appropriate"
                exit_leaf = "01-A-10"
```

Algorithm 1b. UWWTP: Treatment required (Sensitive Areas)

```
if biggestLoad <= 10000:
    if dcpWaterBodyType in ('ES', 'FW', 'LF'):
        result_required = "Secondary"
        alg1_exit_leaf = "01-B-01"
    elif dcpWaterBodyType in ('LC', 'CW'):
        result_required = "Appropriate"
        alg1_exit_leaf = "01-B-02"
else:  # biggestLoad > 10000
    if articleApplies contains '/A54/' or articleApplies contains '/A5854/':
        result_required = "Secondary"
        alg1_exit_leaf = "01-B-03"
    elif articleApplies contains '/A54523/' or articleApplies contains
'/A5854523/':
        result_required = "?"
        alg1_exit_leaf = "01-B-04"
    elif articleApplies contains '/A58523/':
        # Determine result based on treatment parameters and dates
        if not (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
           not (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
           not (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "secondary"
            alg1_exit_leaf = "01-B-05"

        elif not (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(other)"
            alg1_exit_leaf = "01-B-06"

        elif not (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(phosphorus)"
            alg1_exit_leaf = "01-B-07"

        elif not (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(phosphorus+other)"
            alg1_exit_leaf = "01-B-08"

        elif (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(nitrogen)"
            alg1_exit_leaf = "01-B-09"

        elif (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(nitrogen+other)"
            alg1_exit_leaf = "01-B-10"

        elif (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             not (rcaParameterOther and rcaDateArt58 <= repSituationAt):
```

```
            result = "(nitrogen+phosphorus)"
            alg1_exit_leaf = "01-B-11"

        elif (rcaParameterN and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterP and rcaDateArt58 <= repSituationAt) and \
             (rcaParameterOther and rcaDateArt58 <= repSituationAt):
            result = "(nitrogen+phosphorus+other)"
            alg1_exit_leaf = "01-B-12"

        if "/A523/" in article_applies:
            if dcp_type == "CSA":
                goto: algorithm_1c(alg_plant_id)
            elif dcp_type in ["SA", "A523", "A58523"]:
                if rca_a_nitro == 1 and rca_start_date <= rep_situation_at:
                    if not (rca_a_phos == 1 and rca_start_date <=
rep_situation_at) and \

                    not (rca_c == 1 and rca_start_date <= rep_situation_at):
                        result = "(aN)"
                        exit_leaf = "01-B-13"
                    elif not (rca_a_phos == 1) and (rca_c == 1):
                        result = "(aN+c)"
                        exit_leaf = "01-B-14"
                    elif (rca_a_phos == 1) and not (rca_c == 1):
                        result = "(aN+aP)"
                        exit_leaf = "01-B-15"
                    else:
                        result = "(aN+aP+c)"
                        exit_leaf = "01-B-16"

                else:
                    if not (rca_b == 1 and rca_start_date <= rep_situation_at) and
\
                    not (rca_c == 1 and rca_start_date <= rep_situation_at) and \
                    not (rca_a_phos == 1 and rca_start_date <= rep_situation_at):
                        result = "Secondary"
                        exit_leaf = "01-B-17"
                    elif not rca_b and not rca_c and rca_a_phos:
                        result = "(aP)"
                        exit_leaf = "01-B-18"
                    elif not rca_b and rca_c and not rca_a_phos:
                        result = "(c)"
                        exit_leaf = "01-B-19"
                    elif not rca_b and rca_c and rca_a_phos:
                        result = "(aP+c)"
                        exit_leaf = "01-B-20"
                    elif rca_b and not rca_c and not rca_a_phos:
                        result = "(b)"
                        exit_leaf = "01-B-21"
                    elif rca_b and not rca_c and rca_a_phos:
                        result = "(aP+b)"
                        exit_leaf = "01-B-22"
                    elif rca_b and rca_c and not rca_a_phos:
                        result = "(b+c)"
                        exit_leaf = "01-B-23"
                    else:
                        result = "(aP+b+c)"
                        exit_leaf = "01-B-24"
```

Algorithm 1c. UWWTP: Treatment Required (Catchment of Sensitive Areas)

```
if article_5_4_applies:
    result_required = "Secondary"
    exit_leaf = "01-C-07"
else:
    if rcaANitro == 1:
        if rcaAPhos == 1:
            result_required = "(aN+aP)"
            exit_leaf = "01-C-01"
        else:
            result_required = "(aN)"
            exit_leaf = "01-C-02"
    else:
        if rcaB == 0 and rcaAPhos == 0:
            result_required = "Secondary"
            exit_leaf = "01-C-03"
        elif rcaB == 0 and rcaAPhos == 1:
            result_required = "(aP)"
            exit_leaf = "01-C-04"
        elif rcaB == 1 and rcaAPhos == 0:
            result_required = "(b)"
            exit_leaf = "01-C-05"
        elif rcaB == 1 and rcaAPhos == 1:
            result_required = "(aP+b)"
            exit_leaf = "01-C-06"
```

Algorithm 2a. UWWTD: Treatment and Performance Compliance (except more stringent treatment)

```
if dcp_exists == 1:
    if agg_exists_active == 1:
        if active_and_connected == 1:
            if result_required in {"?", "NI"}:
                result_compliance = "?"
                result_treatment = "NR"
                result_performance = "NR"
                exit_leaf = "02-A-03"

            elif result_required in {"Appropriate", "NR"}:
                result_compliance = "NR"
                result_treatment = "not calculable"
                result_performance = "not calculable"
                exit_leaf = "02-A-04"

            elif result_required == "Primary":
                if primary_treatment == 1:
                    result_treatment = "True"
                    if uwwTSSPerf == "P" and uwwBOD5Perf == "P":
                        result_performance = "True"
                        result_compliance = "C"
                        exit_leaf = "02-A-05"
                    else:
                        result_performance = "False"
                        result_compliance = "NC"
                        exit_leaf = "02-A-06"
                else:
                    result_treatment = "False"
                    if uwwTSSPerf == "P" and uwwBOD5Perf == "P":
                        result_performance = "True"
                        result_compliance = "NC"
                        exit_leaf = "02-A-07"
                    else:
                        result_performance = "False"
                        result_compliance = "NC"
                        exit_leaf = "02-A-08"

            elif result_required == "Secondary":
                if secondary_treatment == 1:
                    result_treatment = "True"
                    if uwwCODPerf == "P" and uwwBOD5Perf == "P":
                        result_performance = "True"
                        result_compliance = "C"
                        exit_leaf = "02-A-09"
                    else:
                        result_performance = "False"
                        result_compliance = "NC"
                        exit_leaf = "02-A-10"
                else:
                    result_treatment = "False"
                    if uwwCODPerf == "P" and uwwBOD5Perf == "P":
                        result_performance = "True"
                        result_compliance = "NC"
                        exit_leaf = "02-A-11"
                    else:
```

```
                        result_performance = "False"
                        result_compliance = "NC"
                        exit_leaf = "02-A-12"

            else:
                # More stringent → call Algorithm 2b
                algorithm_2b()

        elif active_and_not_connected == 1:
            result_compliance = "NR"
            result_treatment = "False"
            result_performance = "False"
            exit_leaf = "02-A-15"

        else:
            result_compliance = "NR"
            result_treatment = "not calculable"
            result_performance = "not calculable"
            exit_leaf = "02-A-14"

    else:
        result_compliance = "NR"
        result_treatment = "not calculable"
        result_performance = "not calculable"
        exit_leaf = "02-A-13"

else:
    result_compliance = "?"
    result_treatment = "False"
    result_performance = "False"
    exit_leaf = "02-A-01"

if dcp_exists is None:
    result_compliance = "NI"
    result_treatment = "not calculable"
    result_performance = "not calculable"
    exit_leaf = "02-A-02"
```

Algorithm 2b. UWWTD: Treatment and Performance Compliance (more stringent treatment)

```
# ----------------------------------------------------------------
# CASE 1: Nitrogen + Phosphorus ( (aN+aP) or (nitrogen+phosphorus) )
# ----------------------------------------------------------------
if result_required in ["(aN+aP)", "(nitrogen+phosphorus)"]:
    add_note(f"Required = {result_required}. Need Secondary + N-removal + P-
removal.")

    if secondary_treatment and n_removal and p_removal:
        add_note("Secondary, N-removal, P-removal present.")

        if (
            uwwCODPerf == "P"
            and uwwBOD5Perf == "P"
            and uwwNTotPerf == "P"
            and uwwPTotPerf == "P"
        ):
            add_note("COD, BOD5, N, P all pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-01"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-02"

    else:
        add_note("Missing Secondary or N/P removal.")

        if (
            uwwCODPerf == "P"
            and uwwBOD5Perf == "P"
            and uwwNTotPerf == "P"
            and uwwPTotPerf == "P"
        ):
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-03"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-04"

# ----------------------------------------------------------------
# CASE 2: Phosphorus only ( (aP) or (phosphorus) )
# ----------------------------------------------------------------
elif result_required in ["(aP)", "(phosphorus)"]:
    add_note(f"Required = {result_required}. Need Secondary + P-removal.")
```

```
    if secondary_treatment and p_removal:
        add_note("Secondary + P-removal present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwPTotPerf == "P":
            add_note("COD, BOD5, P pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-05"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-06"

    else:
        add_note("Missing Secondary or P-removal.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwPTotPerf == "P":
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-07"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-08"

# ------------------------------------------------------------------
# CASE 3: Nitrogen only ( (aN) or (nitrogen) )
# ------------------------------------------------------------------
elif result_required in ["(aN)", "(nitrogen)"]:
    add_note(f"Required = {result_required}. Need Secondary + N-removal.")

    if secondary_treatment and n_removal:
        add_note("Secondary + N-removal present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwNTotPerf == "P":
            add_note("COD, BOD5, N pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-09"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-10"

    else:
        add_note("Missing Secondary or N-removal.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwNTotPerf == "P":
```

```
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-11"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-12"

# ------------------------------------------------------------------
# CASE 4: Other treatment ( (c) or (other) )
# ------------------------------------------------------------------
elif result_required in ["(c)", "(other)"]:
    add_note(f"Required = {result_required}. Need Secondary + Other treatment
flag.")

    if secondary_treatment and other_stringent_treatment:
        add_note("Secondary + Other treatment present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P":
            add_note("COD, BOD5 pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-13"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-14"

    else:
        add_note("Missing Secondary or Other treatment.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P":
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-15"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-16"

# ------------------------------------------------------------------
# CASE 5: More Stringent Nitrogen and Phosphorus (aP + b)
# ------------------------------------------------------------------
elif result_required == "(aP+b)":
    add_note(f"Required = {result_required}. Need Secondary + N-removal + P-
removal.")

    if secondary_treatment and n_removal and p_removal:
```

```
        add_note("Secondary, N-removal, P-removal present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwPTotPerf == "P":
            add_note("COD, BOD5, P pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-17"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-18"

    else:
        add_note("Missing Secondary or N/P removal.")
        result_treatment = "False"
        result_performance = "True"
        result_compliance = "NC"
        alg2_exit_leaf = "02-B-19"

# ------------------------------------------------------------------
# CASE 6: Nitrogen + Phosphorus + Other ( (aN+aP+c) or (nitrogen+phosphorus+other)
)
# ------------------------------------------------------------------
elif result_required in ["(aN+aP+c)", "(nitrogen+phosphorus+other)"]:
    add_note(
        f"Required = {result_required}. Need Secondary + N-removal + P-removal +
(any advanced)."
    )

    if secondary_treatment and n_removal and p_removal and advanced_any:
        add_note("All required treatments present.")

        if (
            uwwCODPerf == "P"
            and uwwBOD5Perf == "P"
            and uwwNTotPerf == "P"
            and uwwPTotPerf == "P"
        ):
            add_note("COD, BOD5, N, P pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-20"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-21"

    else:
        add_note("Missing treatment(s).")

        if (
            uwwCODPerf == "P"
            and uwwBOD5Perf == "P"
```

```
            and uwwNTotPerf == "P"
            and uwwPTotPerf == "P"
        ):
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-22"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-23"

# ----------------------------------------------------------------
# CASE 7: Nitrogen + Other ( (aN+c) or (nitrogen+other) )
# ----------------------------------------------------------------
elif result_required in ["(aN+c)", "(nitrogen+other)"]:
    add_note(
        f"Required = {result_required}. Need Secondary + N-removal + (any
advanced)."
    )

    if secondary_treatment and n_removal and advanced_any:
        add_note("Secondary + N-removal + advanced present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwNTotPerf == "P":
            add_note("COD, BOD5, N pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-24"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-25"

    else:
        add_note("Missing treatment(s).")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwNTotPerf == "P":
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-26"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-27"

# ----------------------------------------------------------------
# CASE 8: Phosphorus + Other ( (aP+c) or (phosphorus+other) )
# ----------------------------------------------------------------
```

```
elif result_required in ["(aP+c)", "(phosphorus+other)"]:
    add_note(
        f"Required = {result_required}. Need Secondary + P-removal + (any
advanced)."
    )

    if secondary_treatment and p_removal and advanced_any:
        add_note("Secondary + P-removal + advanced present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwPTotPerf == "P":
            add_note("COD, BOD5, P pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-28"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-29"

    else:
        add_note("Missing treatment(s).")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P" and uwwPTotPerf == "P":
            add_note("Performance ok but treatment missing → NC.")
            result_treatment = "False"
            result_performance = "True"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-30"
        else:
            add_note("Performance + treatment missing → NC.")
            result_treatment = "False"
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-31"

# ------------------------------------------------------------------
# CASE 9: N-removal + Advanced ( (b+c) )
# ------------------------------------------------------------------
elif result_required == "(b+c)":
    add_note(
        f"Required = {result_required}. Need Secondary + N-removal + (any
advanced)."
    )

    if secondary_treatment and n_removal and advanced_any:
        add_note("Secondary + N-removal + advanced present.")

        if uwwCODPerf == "P" and uwwBOD5Perf == "P":
            add_note("COD, BOD5 pass → Compliant.")
            result_treatment = "True"
            result_performance = "True"
            result_compliance = "C"
            alg2_exit_leaf = "02-B-32"
        else:
            add_note("Performance failed → Non-compliant.")
            result_treatment = "True"
```

```
            result_performance = "False"
            result_compliance = "NC"
            alg2_exit_leaf = "02-B-33"

    else:
        add_note("Missing treatment(s).")
        result_treatment = "False"
        result_performance = "False"
        result_compliance = "NC"
        alg2_exit_leaf = "02-B-34"
```

Algorithm 3. UWWTD: Treatment and Performance Compliance (Correction for Transitional Period)

```
# --- Case 1: Required = Appropriate ---
if required == "Appropriate":
    if deadline_art4 is None or deadline_art4 <= rep_date:
        log("Deadline art.4 passed, required=Appropriate → Compliance = Algorithm
2 result")
        alg3_exit_leaf = "03-01"
        # compliance stays = compliance_alg2
    else:
        log("Deadline art.4 not yet reached, required=Appropriate → Compliance =
PD")
        compliance_alg2 = "PD"
        alg3_exit_leaf = "03-02"

# --- Case 2: Required = Primary / Secondary / More stringent ---
elif required in [
    "Primary","Secondary",
    "(aN)","(aN+aP)","(aN+aP+c)","(aN+c)",
    "(aP)","(aP+b)","(aP+b+c)","(aP+c)",
    "(b)","(b+c)","(c)",
    "(nitrogen)","(nitrogen+other)",
    "(nitrogen+phosphorus)","(nitrogen+phosphorus+other)",
    "(other)","(phosphorus)","(phosphorus+other)"
]:
    if deadline_art4 is None or deadline_art4 <= rep_date:
        log("Deadline art.4 passed → check art.5 deadline")

        if deadline_art5 is None or deadline_art5 <= rep_date:
            log("Deadline art.5 passed → Compliance = Algorithm 2 result")
            alg3_exit_leaf = "03-04"

        else:
            log("Deadline art.5 not passed → check secondary treatment + COD +
BOD5")
            if has_secondary and bod5_perf == "P" and cod_perf == "P":
                log("Secondary present & COD+BOD5 passed → Compliance = C")
                compliance_alg2 = "C"
                alg3_exit_leaf = "03-05"
            else:
                log("Secondary missing or COD/BOD5 failed → Compliance = NC")
                compliance_alg2 = "NC"
                alg3_exit_leaf = "03-06"
    else:
        log("Deadline art.4 not yet reached → Compliance = PD")
        compliance_alg2 = "PD"
        alg3_exit_leaf = "03-03"
```

Algorithm 4. Agglomeration: Article 3 Compliance

```
if aggGenerated < 2000:
    setCompliance(agg_id, art3="NR", art4="NR", art5="NR", art6="NR", exit="04-01")

else:
    if aggDateArt3 <= repSituationAt:
        # deadline reached
        untreated_load = (aggGenerated * aggPercWithoutTreatment) / 100
        ias_load = (aggGenerated * aggC2) / 100

        if untreated_load + ias_load <= 2000 and (aggPercWithoutTreatment + aggC2) <= 2:
            setCompliance(agg_id, art3="C", exit="04-02")

        elif untreated_load > 2000 or aggPercWithoutTreatment > 2:
            setCompliance(agg_id, art3="NC", exit="04-05")

        else:
            # borderline, check IAS
            if aggC2 > 2 or ias_load > 1000:
                setCompliance(agg_id, art3="C", add_art3="AddQC", exit="04-06")
            else:
                setCompliance(agg_id, art3="C", add_art3="QC", exit="04-03")

    else:
        # deadline not reached
        setCompliance(agg_id, art3="PD", exit="04-04")
```

Algorithm 5. Agglomeration: Article 4, 5, 6 Compliance Exclusion

```
if sum_wastewater_collected > 0:
    if num_plants_connected > 0:
        # Check if all connected plants are "Appropriate" & NR/NI
        if num_plants_required_appropriate_nr_ni == num_plants_connected:
            setCompliance(agg_id, art4="NR", art5="NR", art6="NR", exit="05-04")
            algorithm10(agg_id)
            algorithm11(agg_id)
            logFinalResults(agg_id)
        else:
            # Check if at least one "?" and others C/NR/PD
            if (
                num_plants_comp_int > 0
                and (num_plants_comp_int + num_plants_comp_c_nr_pd)
                == num_plants_connected
            ):
                setCompliance(agg_id, art4="?", art5="?", art6="?", exit="05-02")
            else:
                algorithm6(agg_id)
    else:
        # No plants connected
        if aggGenerated > 10000:
            setCompliance(agg_id, art4="?", art5="?", art6="NR", exit="05-01")
        else:
            setCompliance(agg_id, art4="?", art5="NR", art6="NR", exit="05-03")
        algorithm6(agg_id)
else:
    # No wastewater collected
    algorithm6(agg_id)
```

Algorithm 6. Agglomeration: UWWTP Compliance for Article 4, 5, and 6

```
# Determine if a more stringent treatment is required
stringent_treatments = [
    '(aN)', '(aN+aP)', '(aN+aP+c)', '(aN+c)', '(aP)', '(aP+b)', '(aP+b+c)',
'(aP+c)', '(b)', '(b+c)', '(c)', '(nitrogen)', '(nitrogen+other)',
'(nitrogen+phosphorus)', '(nitrogen+phosphorus+other)',
'(other)',    '(phosphorus)', '(phosphorus+other)']


# --- Main decision tree ---
if sum_percent_wastewater_collected + discharge_without_treatment == 0 or
result_required == 'NR':
    # Case 06-01
    add_note(alg_agglo_id, 4, "Collected wastewater + discharge_without_treatment
= 0 OR Treatment required = NR => Compliance = NR for Art.4, Art.5, Art.6")
    update_plant_compliance(alg_plant_id, art4='NR', art5='NR', art6='NR',
exit_leaf='06-01')

elif result_required == 'Primary':
    add_note(alg_agglo_id, 4, "Treatment required = Primary => Compliance has to
be checked")
        add_note(alg_agglo_id, 4, "Country not Portugal => Compliance = NR for
Art.4, Art.5, Art.6")
        update_plant_compliance(alg_plant_id, art4='NR', art5='NR', art6='NR',
exit_leaf='06-21')
        if result_compliance == 'C':
            add_note(alg_agglo_id, 4, "Compliance = 'C' => Art.6 = C, Art.4/5 =
NR")
            update_plant_compliance(alg_plant_id, art4='NR', art5='NR', art6='C',
exit_leaf='06-02')
        else:
            if aggDateArt3 <= rep_situation_at:
                add_note(alg_agglo_id, 4, "Deadline art.6 <= Reporting year =>
Art.6 = NC, Art.4/5 = NR")
                update_plant_compliance(alg_plant_id, art4='NR', art5='NR',
art6='NC', exit_leaf='06-03')
            else:
                add_note(alg_agglo_id, 4, "Deadline art.6 > Reporting year =>
Art.6 = PD, Art.4/5 = NR")
                update_plant_compliance(alg_plant_id, art4='NR', art5='NR',
art6='PD', exit_leaf='06-04')

else:  # Treatment not Primary
    add_note(alg_agglo_id, 4, "Treatment required not Primary => Treatment and
treatment in place have to be checked")

    # Secondary treatment required and primary treatment in place
    if result_required == 'Secondary' and highest_treatment_in_place == '1':
        if aggDateArt4 <= rep_situation_at:
            update_plant_compliance(alg_plant_id, art4='NC', art5='NR', art6='NR',
exit_leaf='06-05')
        else:
            update_plant_compliance(alg_plant_id, art4='PD', art5='NR', art6='NR',
exit_leaf='06-06')

    # Secondary treatment required and secondary or stringent treatment in place
    elif result_required == 'Secondary' and highest_treatment_in_place in ['2',
'S']:
```

```
        if uwwCODPerf == 'P' and uwwBOD5Perf == 'P':
            if aggDateArt5 <= rep_situation_at:
                update_plant_compliance(alg_plant_id, art4='C', art5='NR',
art6='NR', exit_leaf='06-07')
            else:
                update_plant_compliance(alg_plant_id, art4='C', art5='PD',
art6='NR', exit_leaf='06-21')
        else:
            if aggDateArt4 <= rep_situation_at:
                if aggDateArt5 <= rep_situation_at:
                    update_plant_compliance(alg_plant_id, art4='NC', art5='NC',
art6='NR', exit_leaf='06-18')
                else:
                    update_plant_compliance(alg_plant_id, art4='NC', art5='PD',
art6='NR', exit_leaf='06-19')
            else:
                update_plant_compliance(alg_plant_id, art4='PD', art5='PD',
art6='NR', exit_leaf='06-20')

    # More stringent treatment required
    elif required_more_stringent and result_treatment == 'NR':
        if aggDateArt4 <= rep_situation_at:
            if aggDateArt5 <= rep_situation_at:
                update_plant_compliance(alg_plant_id, art4='NC', art5='NC',
art6='NR', exit_leaf='06-13')
            else:
                update_plant_compliance(alg_plant_id, art4='NC', art5='PD',
art6='NR', exit_leaf='06-14')
        else:
            update_plant_compliance(alg_plant_id, art4='PD', art5='PD', art6='NR',
exit_leaf='06-12')

    # COD/BOD check fallback
    else:
        if uwwCODPerf == 'P' and uwwBOD5Perf == 'P':
            if result_compliance == 'C':
                update_plant_compliance(alg_plant_id, art4='C', art5='C',
art6='NR', exit_leaf='06-15')
            else:
                if aggDateArt5 <= rep_situation_at:
                    update_plant_compliance(alg_plant_id, art4='C', art5='NC',
art6='NR', exit_leaf='06-16')
                else:
                    update_plant_compliance(alg_plant_id, art4='C', art5='PD',
art6='NR', exit_leaf='06-17')
        else:
            if aggDateArt4 <= rep_situation_at:
                if aggDateArt5 <= rep_situation_at:
                    update_plant_compliance(alg_plant_id, art4='NC', art5='NC',
art6='NR', exit_leaf='06-18')
                else:
                    update_plant_compliance(alg_plant_id, art4='NC', art5='PD',
art6='NR', exit_leaf='06-19')
            else:
                update_plant_compliance(alg_plant_id, art4='PD', art5='PD',
art6='NR', exit_leaf='06-20')
```

Algorithm 7. Agglomeration: Article 4 only

```
A = (aggC1 * aggGenerated) * 0.01

if np_treat_second_more_str > 0:
    # At least one station requires secondary or more stringent treatment
    if (
        aggPercWithoutTreatment <= 2.0
        and (aggGenerated * aggPercWithoutTreatment) / 100 <= 2000
    ):
        # Load discharged without treatment is low
        if (
            A - (sum_percent_wastewater_collected * aggGenerated) * 0.01
        ) + sp_load_enter_treatNR_4NC <= A * 0.01 or (
            (
                aggC1 * aggGenerated * 0.01
                - sum_percent_wastewater_collected * aggGenerated * 0.01
            )
            + sp_load_enter_treatNR_4NC
            < 2000
        ):
            if (
                np_4NC > 0 and sp_load_enter_4NC > A * 0.01
            ) or sp_load_enter_4NC >= 2000:
                # Deadline check for Article 4
                if aggDateArt4 <= repSituationAt:
                    return "NC", "07-04"
                else:
                    return "PD", "07-05"
            else:
                # Check PD or C plants
                if np_4PD > 0 or aggDateArt4 > repSituationAt:
                    return "PD", "07-06"
                elif np_4C > 0:
                    return "C", "07-07"
                else:
                    return "NR", "07-08"
        else:
            # Load condition not met → check deadlines
            if aggDateArt4:
                if aggDateArt4 <= repSituationAt:
                    return "NC", "07-10"
                else:
                    return "PD", "07-11"
            else:
                return "NR", "07-09"
    else:
        # Load discharged without treatment > threshold → check deadline
        if aggDateArt4 <= repSituationAt:
            return "NC", "07-03"
        else:
            return "PD", "07-02"
else:
    # No secondary or more stringent treatment required
    return "NR", "07-01"
```

Algorithm 8. Agglomeration: Article 6 only

```
A = (aggC1 * aggGenerated) * 0.01

if np_treat_primary > 0:
    # At least one primary treatment plant
    if np_6NC > 0 and sp_load_enter_6NC > A*0.01 and sp_load_enter_6NC >= 2000:
        return "NC", "08-02"
    elif np_6PD > 0:
        return "PD", "08-03"
    elif np_6C > 0:
        return "C", "08-04"
    else:
        return "NR", "08-05"
else:
    # No primary treatment plant
    return "NR", "08-01"
```

Algorithm 9. Agglomeration: Article 5 only

```
A = (aggC1 * aggGenerated) * 0.01

if aggGenerated < 10000:
    return "NR", "09-01"

if np_54 > 0:
    if remN_remP_75 > 0:
        return "C", "09-02"
    else:
        return "NC", "09-13"

if np_treat_more_str == 0:
    return "NR", "09-03"

# Check load discharged without treatment
if (
    aggPercWithoutTreatment <= 2.0
    and (aggGenerated * aggPercWithoutTreatment * 0.01) <= 2000
):
    if np_5NC > 0 and sp_load_enter_5NC > A * 0.01 and sp_load_enter_5NC >=
2000:
        return "NC", "09-06"
    elif np_5PD > 0:
        return "PD", "09-07"
    elif np_5C > 0:
        return "C", "09-08"
    else:
        return "NR", "09-09"
else:
    # Check Article 5 deadline
    if aggDateArt5 is None:
        return "NR", "09-10"
    elif aggDateArt5 <= repSituationAt:
        return "NC", "09-04"  # or 09-11 depending on branch
    else:
        return "PD", "09-05"  # or 09-12 depending on branch
```

Algorithm 10. Agglomeration: Overall Compliance

```
# Unknown compliance
if "?" in (art3, art4, art5, art6):
    return "?", "10-01"

# If any NC
if "NC" in (art3, art4, art5, art6):
    return "NC", "10-02"

# Hierarchical checks
if art3 == "NR":
    return "NR", "10-03"
elif art3 == "PD":
    return "PD", "10-04"
elif art3 not in ("C", "QC"):
    return "?", "10-05"
elif aggGenerated < 10000 and np_treat_appropriate > 0:
    return "C", "10-06"
elif art6 in ("C", "PD"):
    return "C", "10-07"
elif art4 in ("C", "PD", "NR"):
    return "C", "10-08"
else:
    return "NR", "10-09"
```

Algorithm 11. Agglomeration: Legal Compliance of Articles 4, 5, and 6

```
# Article 3 = NC branch
if art3 == "NC":
    if art6 not in ("NR", "PD"):
        return ("NR", "NR", "NC"), "11-02"
    else:
        if art4 not in ("NR", "PD"):
            if art5 not in ("NR", "PD"):
                return ("NC", "NC", "NR"), "11-04"
            else:
                return ("NC", art5, "NR"), "11-05"
        else:
            return (art4, art5, art6), "11-03"

# Article 3 != NC branch
else:
    if np_54 > 0:
        if aggGenerated < 10000:
            return (art4, "NR", art6), "11-08"
        else:
            if art4 == "C":
                return (art4, "NR", art6), "11-06"
            else:
                return (art4, "NC", art6), "11-07"
    else:
        if art4 == "NC":
            if art5 not in ("NR", "PD"):
                return ("NC", "NC", "NR"), "11-04"
            else:
                return ("NC", art5, "NR"), "11-05"
        else:
            return (art4, art5, art6), "11-01"
```